

Export atlas

every image and table the page offers for download,
shown once as it looks on a sound run and once on a deviating one

57 of 99 catalogue entries are exercised by the demonstration data and appear with a pair of figures.
42 entries are listed at the end with the page's own reason for showing nothing.

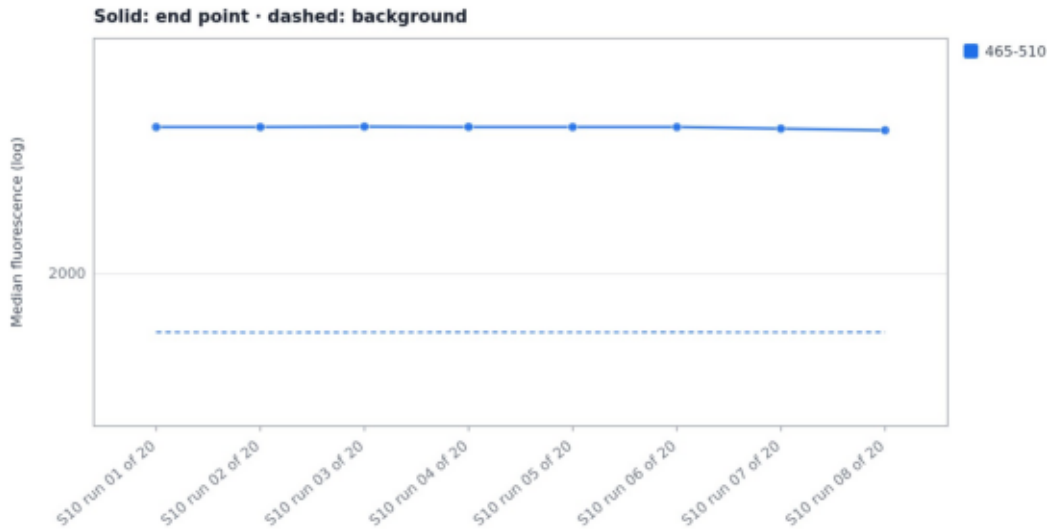
Every figure was produced by loading generated experiment files into the page itself and exporting through its own download path.
No value in this document was measured.

RUO NOT VALIDATED

img:x_signal · Across runs · figure

SOUND S10_run01.ix, S10_run02.ix (+6 more)

DEVIATION S10_run19.ix, S10_run20.ix (8 files)



WHAT IT SHOWS

Median background and median end-point fluorescence of every channel per run: slow drift here is an instrument or reagent trend.

WHAT CHANGED AND WHAT IT MEANS

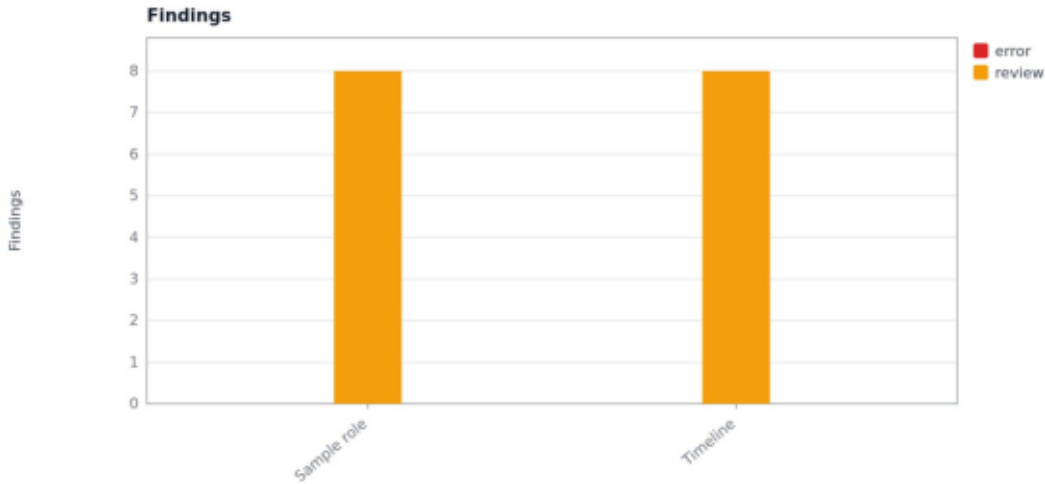
49 % of everything drawn on the figure moved between the two sides. The deviating side is the first eight runs of the series beside its last eight of S10, efficiency falling over the series. Onset: run 06. From run 6 the calibration spacing and amplitude deteriorate; stored efficiency falls from about 1.01 toward 0.76 by run 20. What to look at: The first persistent downward change is run 6; later runs widen the separation from baseline.

also moved by: S1 (37 %), S25 (36 %), S23 (33 %), S22 (29 %), S50 (29 %)

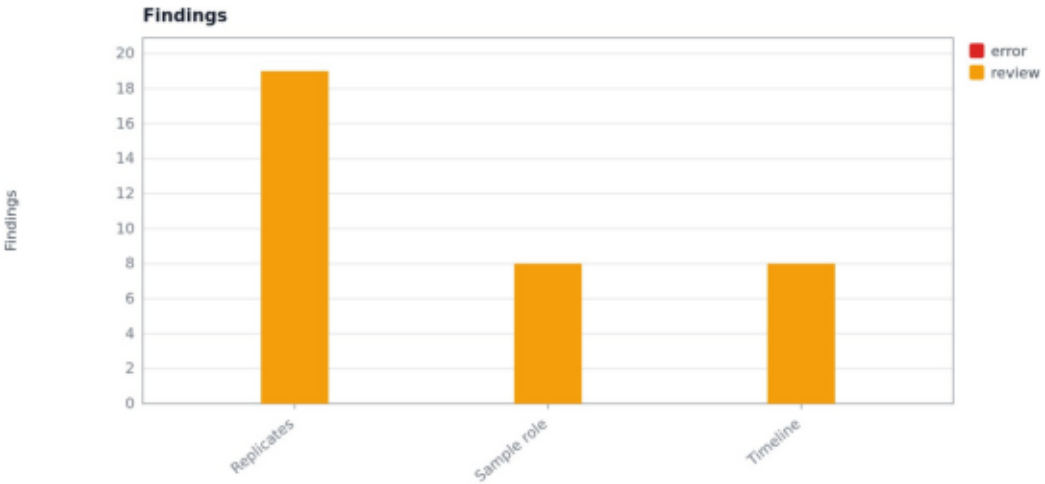
RUO NOT VALIDATED

img:x_findings · Across runs · figure

SOUND S19_run01.eds, S19_run02.eds (+6 more)



DEVIATION S19_run19.eds, S19_run20.eds (8 files)



WHAT IT SHOWS

Count of error and review findings in the forensic log, grouped by area, largest first.

WHAT CHANGED AND WHAT IT MEANS

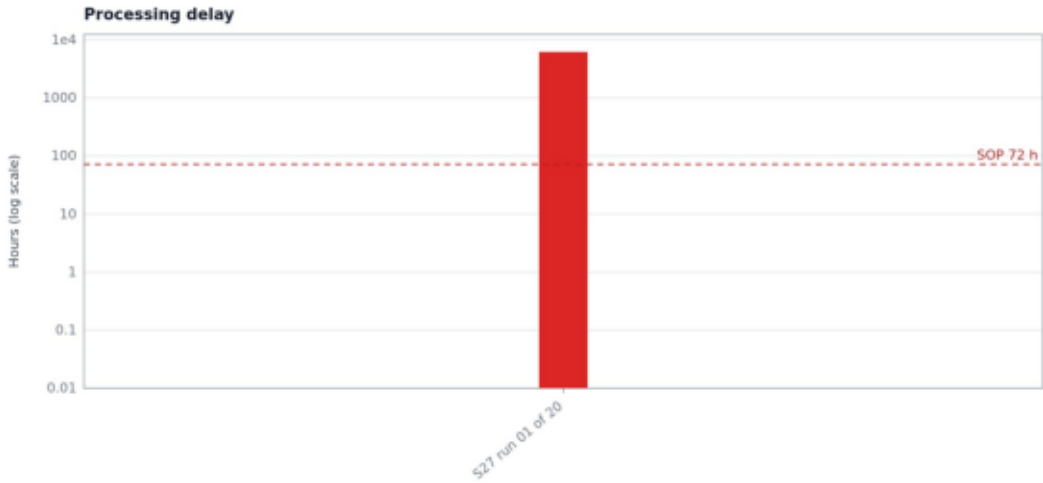
95 % of everything drawn on the figure moved between the two sides. The deviating side is the first eight runs of the series beside its last eight of S19, passive reference falling away. Onset: run 05. ROX declines from run 5 onward to about 55% below baseline by run 20; ROXLOW/BADROX flags and replicate spread increase with it. What to look at: The passive-reference trend starts before the spread worsens; use ROX and flags as leading evidence.

also moved by: S19 (94 %), S28 (92 %), S28 (92 %), S5 (90 %), S36 (76 %)

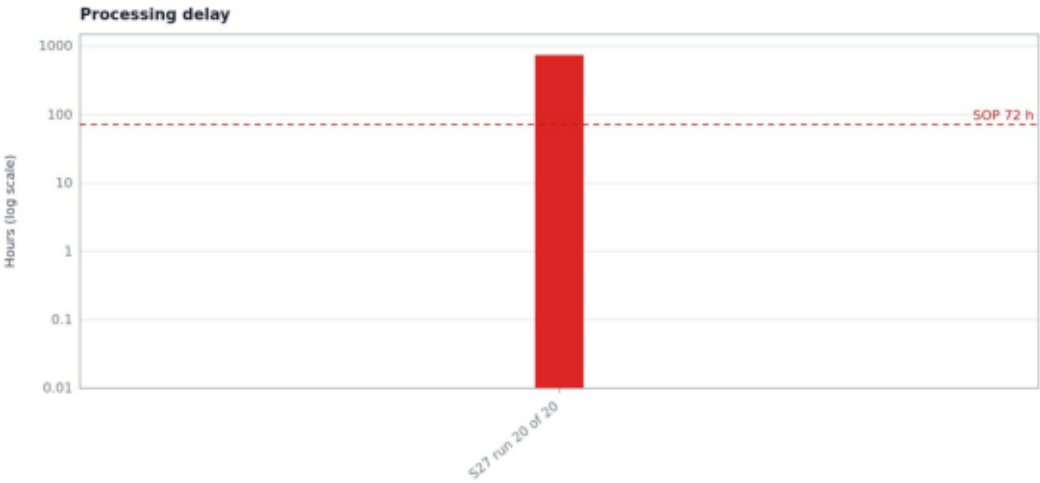
RUO NOT VALIDATED

img:x_delay · Across runs · figure

SOUND S27_run01.eds



DEVIATION S27_run20.eds



WHAT IT SHOWS

Processing time per run. The dashed line is the SOP maximum; bars above it are flagged in the run acceptance.

WHAT CHANGED AND WHAT IT MEANS

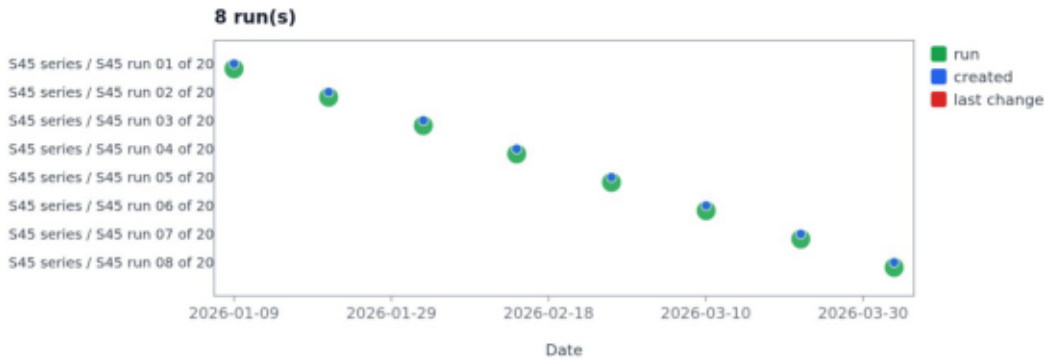
34 % of everything drawn on the figure moved between the two sides. The deviating side is the first run of the series beside its last run of S27, reagent lot change. Onset: run 08. Master-mix lot changes from A to B at run 8 and every Cq shifts about +0.5 cycles. What to look at: The run-08 boundary is a synchronized step across targets and controls.

also moved by: S21 (34 %), S22 (34 %), S19 (34 %), S23 (34 %), S25 (34 %)

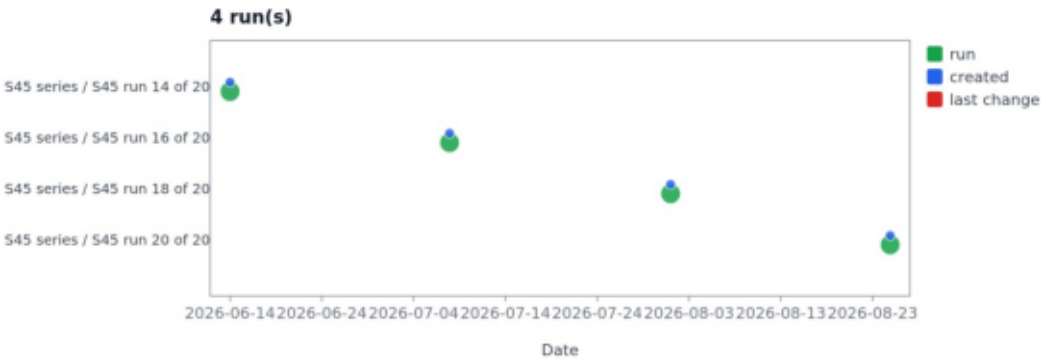
RUO NOT VALIDATED

img:x_timeline · Across runs · figure

SOUND S45_run01.rdm1, S45_run02.rdm1 (+6 more)



DEVIATION S45_run19.rdm1, S45_run20.rdm1 (8 files)



WHAT IT SHOWS

Each run as a green dot at its start, with the experiment creation (blue, above) and the last recorded change (red, below, joined by a dashed line). Long gaps between run and last change are the first thing an auditor asks about.

WHAT CHANGED AND WHAT IT MEANS

74 % of everything drawn on the figure moved between the two sides. The deviating side is the first eight runs of the series beside its last eight of S45, run date missing from selected files. Onset: runs 13, 15, 17 and 19. Four intermittent files omit `runDate`; the series cannot be ordered from file metadata alone at those points. What to look at: Missing dates recur at odd-numbered later runs rather than forming one continuous block.

also moved by: S8 (67 %), S3 (66 %), S5 (66 %), S1 (65 %), S4 (64 %)

RUO

NOT VALIDATED

cc:exposure

Control charts

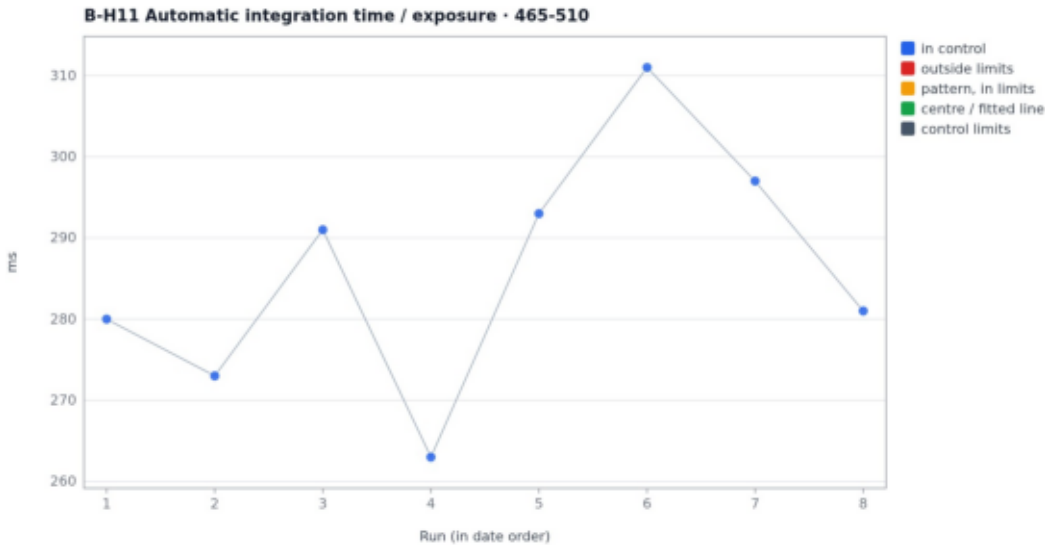
hardware

figure

SOUND

S9_run01.ixx, S9_run02.ixx

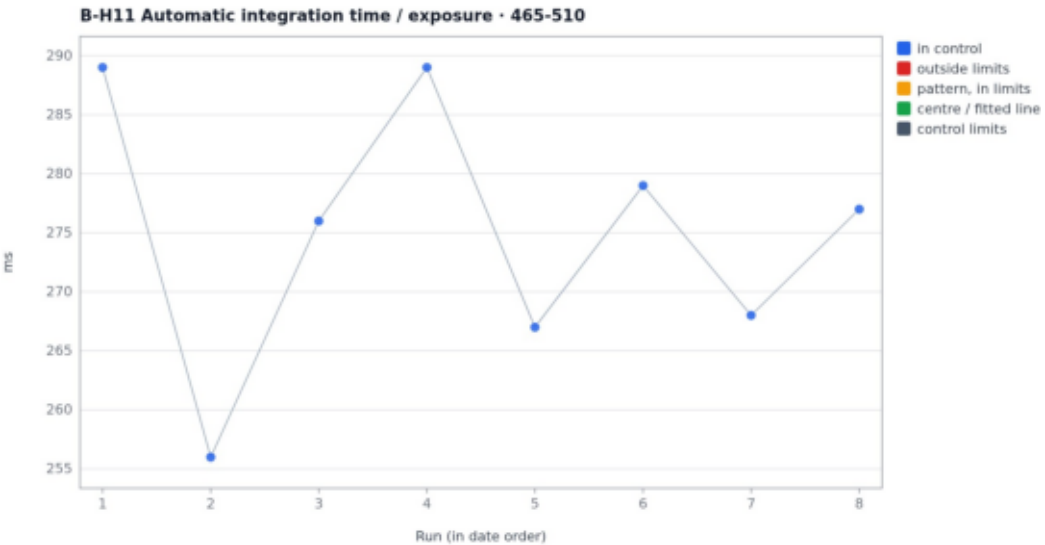
(+6 more)



DEVIATION

S9_run19.ixx, S9_run20.ixx

(8 files)



WHAT IT SHOWS

A slow rise is compatible with weaker signal; verify matched controls and optical configuration before attributing ageing (compare with B-H8/B-H10); a jump after a reagent change points to the method instead.

HOW TO READ THE TWO SIDES

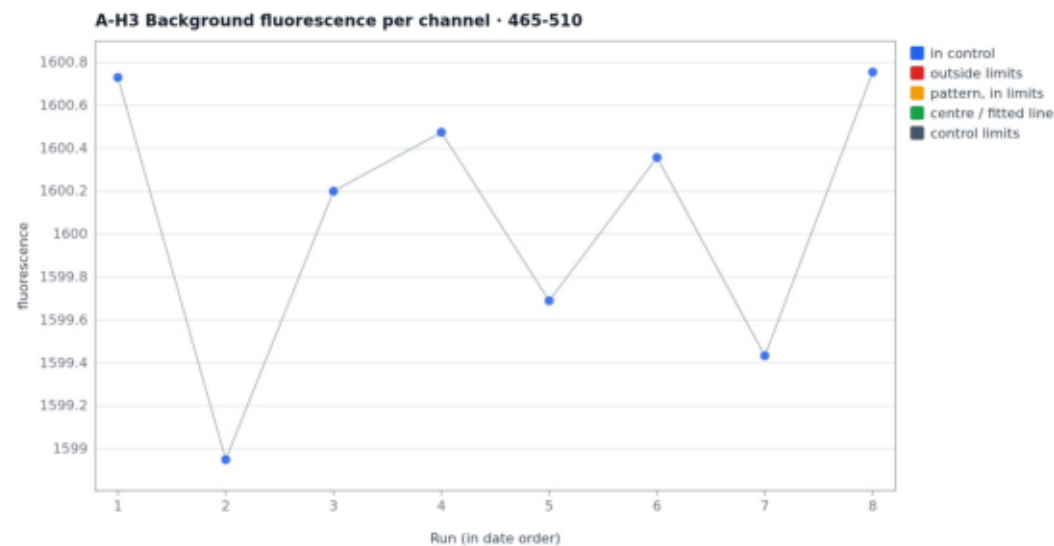
64 % of everything drawn moved between the two sides. The runs come from S9, calibrator shift between lots — the first eight runs of the series against its last eight. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. What matters is the level: the corpus of laboratory and vendor files puts it between 170 and 939 ms, around 284, and the generated history is written to sit inside that.

also moved by: S10 (61 %), S16 (60 %), S11 (59 %), S13 (59 %), S9 (59 %)

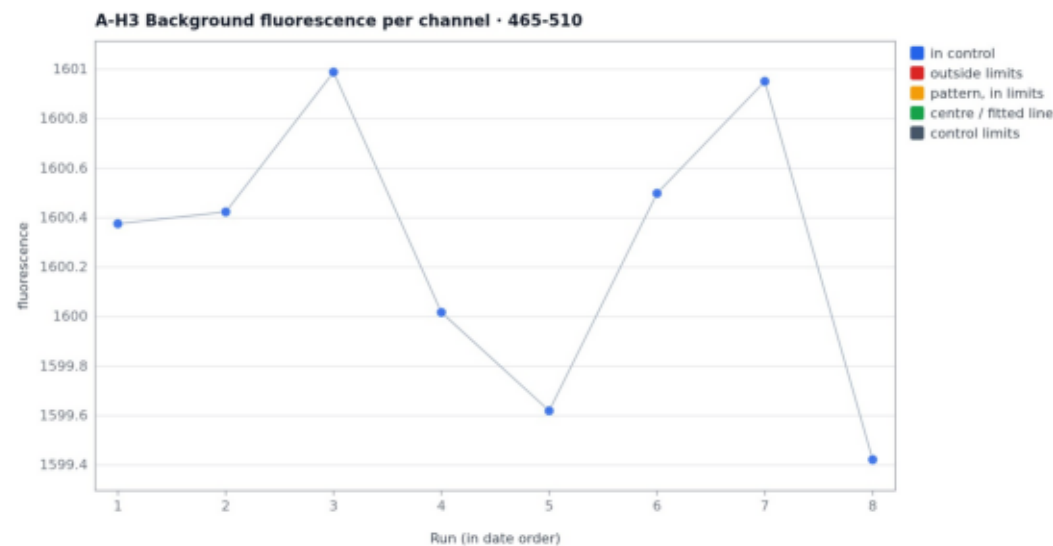
RUO NOT VALIDATED

cc:bg_median · Control charts · hardware · figure

SOUND S10_run01.ixo, S10_run02.ixo (+6 more)



DEVIATION S10_run19.ixo, S10_run20.ixo (8 files)



WHAT IT SHOWS

Slow decline = light source ageing; a raised run = plate, seal or contamination.

HOW TO READ THE TWO SIDES

68 % of everything drawn moved between the two sides. The runs come from S10, efficiency falling over the series — the first eight runs of the series against its last eight. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. What matters is the level: the corpus of laboratory and vendor files puts it between 1427 and 2002 fluorescence, around 1586, and the generated history is written to sit inside that.

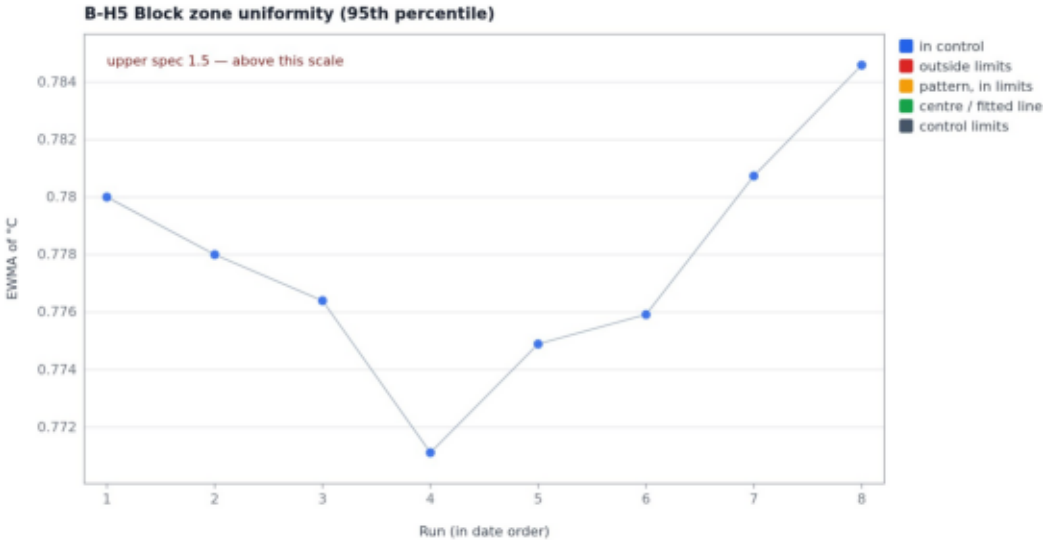
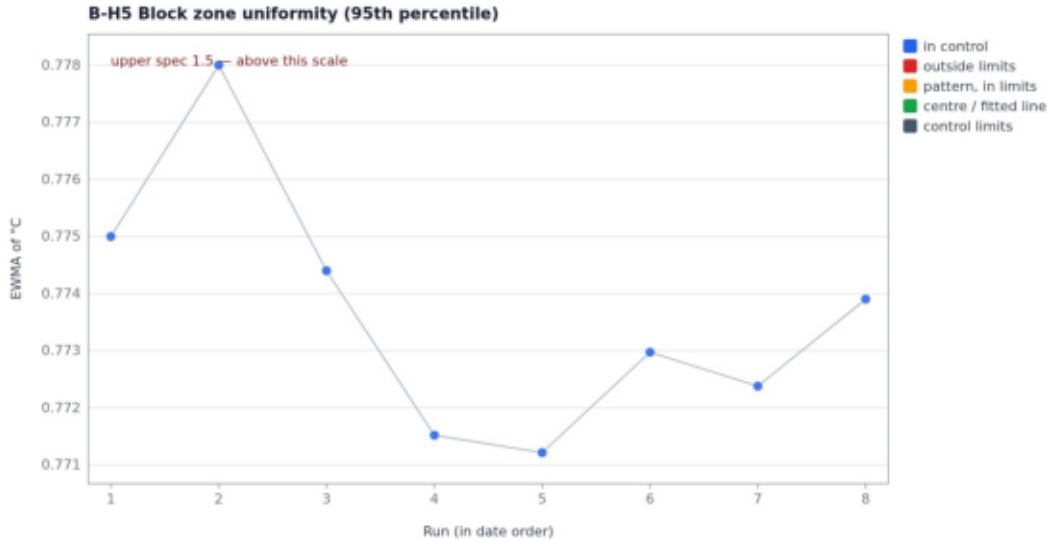
also moved by: S27 (65 %), S38 (65 %), S47 (64 %), S23 (64 %), S31 (63 %)

RUO NOT VALIDATED

cc:zone_spread · Control charts · hardware · figure

SOUND S27_run01.eds, S27_run02.eds (+6 more)

DEVIATION S27_run19.eds, S27_run20.eds (8 files)



WHAT IT SHOWS

An EWMA creeping up means one zone is losing its match; look at the Temperature graph to find which.

HOW TO READ THE TWO SIDES

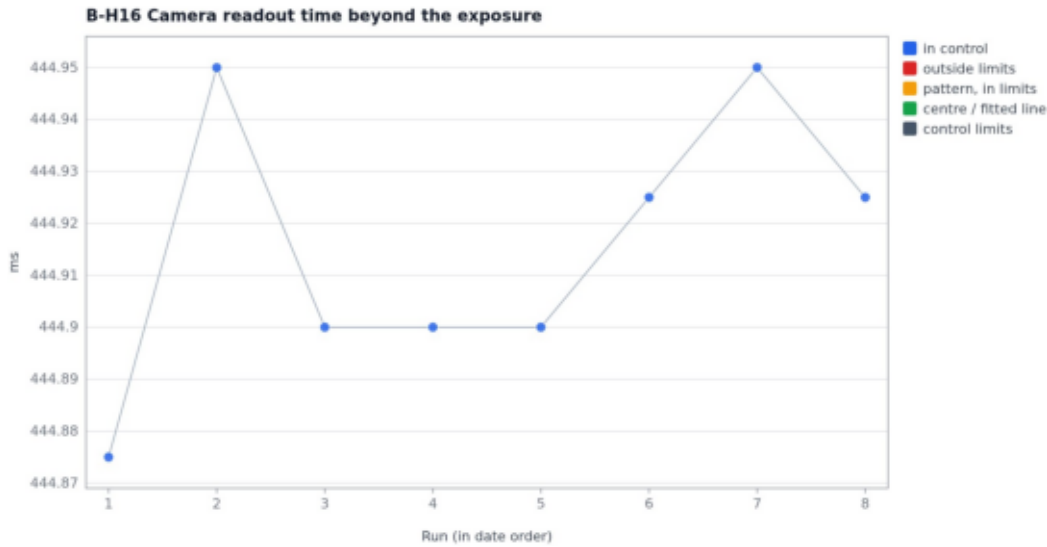
60 % of everything drawn moved between the two sides. The runs come from S27, reagent lot change — the first eight runs of the series against its last eight. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. What matters is the level: the corpus of laboratory and vendor files puts it between 0.5 and 1.5 °C, around 0.5, and the generated history is written to sit inside that.

also moved by: S28 (59 %), S30 (59 %), S23 (59 %), S19 (58 %), S26 (58 %)

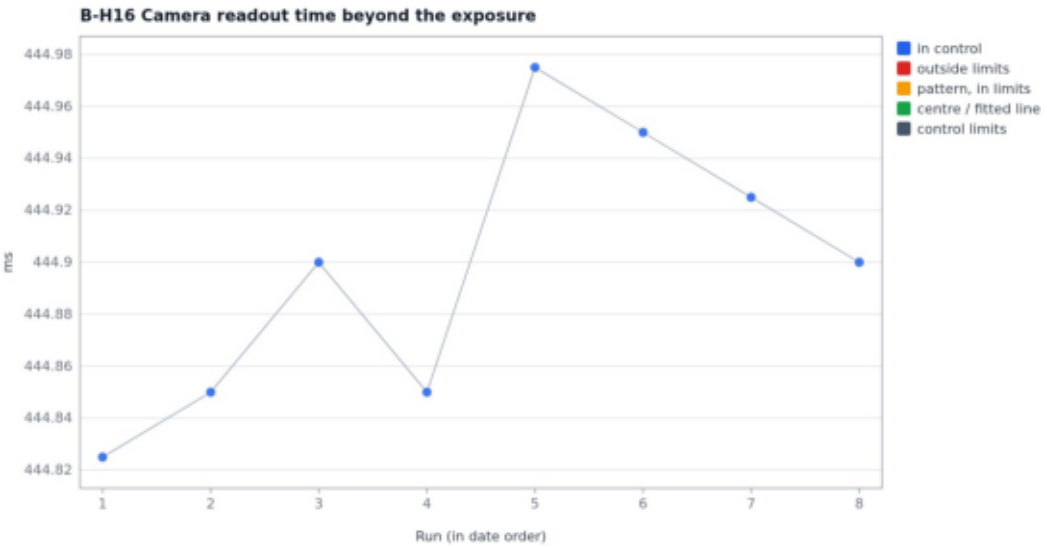
RUO NOT VALIDATED

cc:cam_overhead · Control charts · hardware · figure

SOUND S21_run01.eds, S21_run02.eds (+6 more)



DEVIATION S21_run19.eds, S21_run20.eds (8 files)



WHAT IT SHOWS

A jump = camera or its interface; a slow rise = controller slowing (compare with software charts).

HOW TO READ THE TWO SIDES

67 % of everything drawn moved between the two sides. The runs come from S21, threshold switched from automatic to manual — the first eight runs of the series against its last eight. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. What matters is the level: the corpus of laboratory and vendor files puts it between 432 and 445.3 ms, around 444.9, and the generated history is written to sit inside that.

also moved by: S30 (67 %), S20 (66 %), S24 (65 %), S23 (64 %), S27 (63 %)

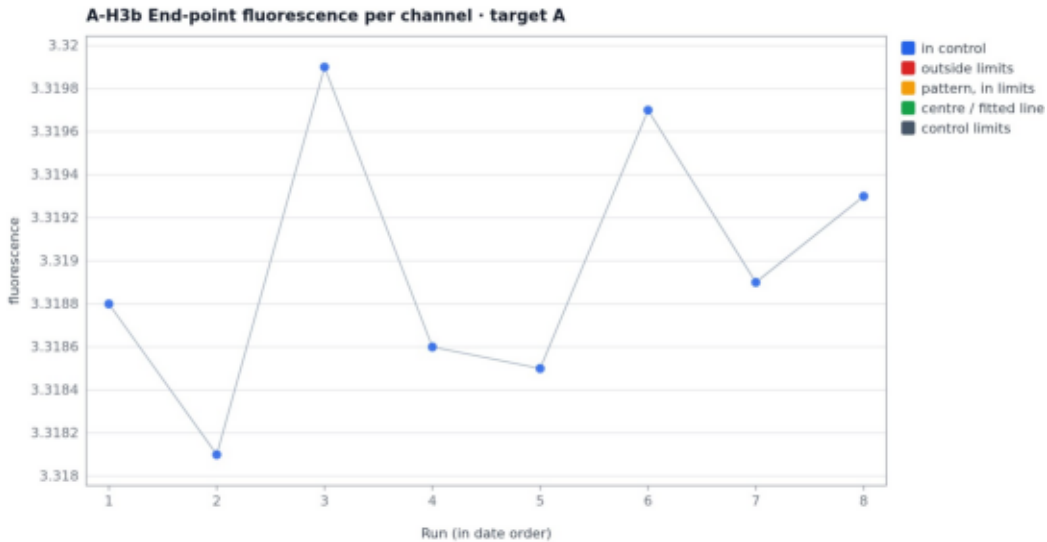
RUO

NOT VALIDATED

cc:plateau_median · Control charts · hardware · figure

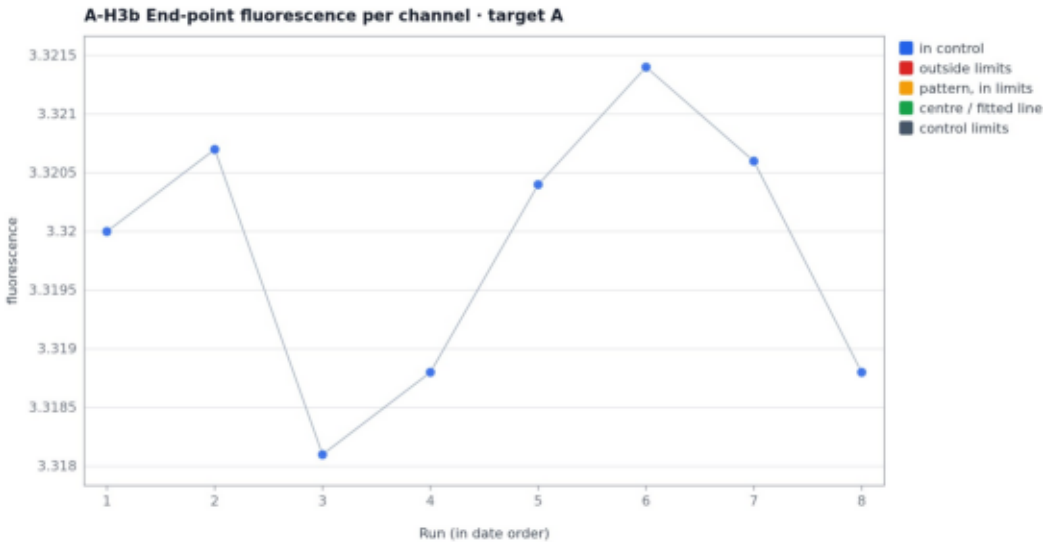
SOUND

S38_run01.rdm1, S38_run02.rdm1 (+6 more)



DEVIATION

S38_run19.rdm1, S38_run20.rdm1 (8 files)



WHAT IT SHOWS

Moves together with B-H8/B-H10 when optics age; alone it points to the method (probe) — see M-group.

HOW TO READ THE TWO SIDES

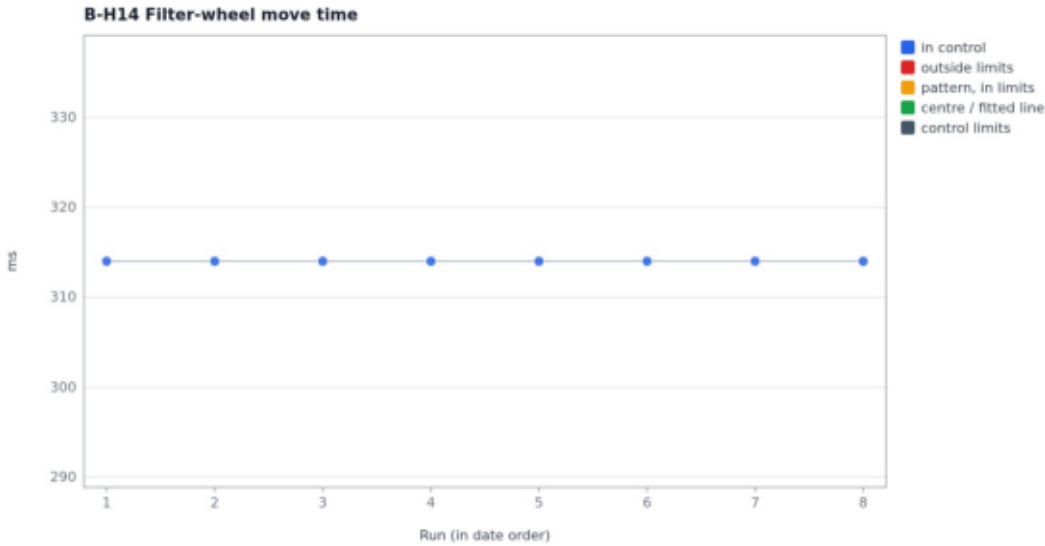
71 % of everything drawn moved between the two sides. The runs come from S38, reference target loses its type — the first eight runs of the series against its last eight. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. What matters is the level: the corpus of laboratory and vendor files puts it between 4.241 and 13.51 fluorescence, around 8.827, and the generated history is written to sit inside that.

also moved by: S45 (65 %), S44 (65 %), S19 (65 %), S40 (63 %), S36 (63 %)

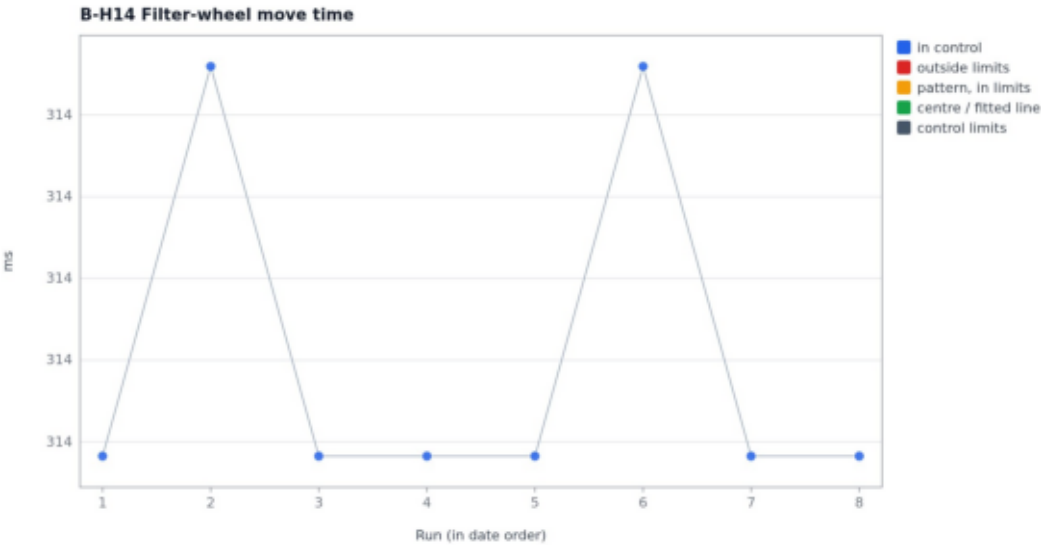
RUO NOT VALIDATED

cc:wheel_ms · Control charts · hardware · figure

SOUND S24_run01.eds, S24_run02.eds (+6 more)



DEVIATION S24_run19.eds, S24_run20.eds (8 files)



WHAT IT SHOWS

Slower moves = motor or bearing wear.

HOW TO READ THE TWO SIDES

60 % of everything drawn moved between the two sides. The runs come from S24, amplification in negative controls — the first eight runs of the series against its last eight. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. What matters is the level: the corpus of laboratory and vendor files puts it between 87 and 315 ms, around 314, and the generated history is written to sit inside that.

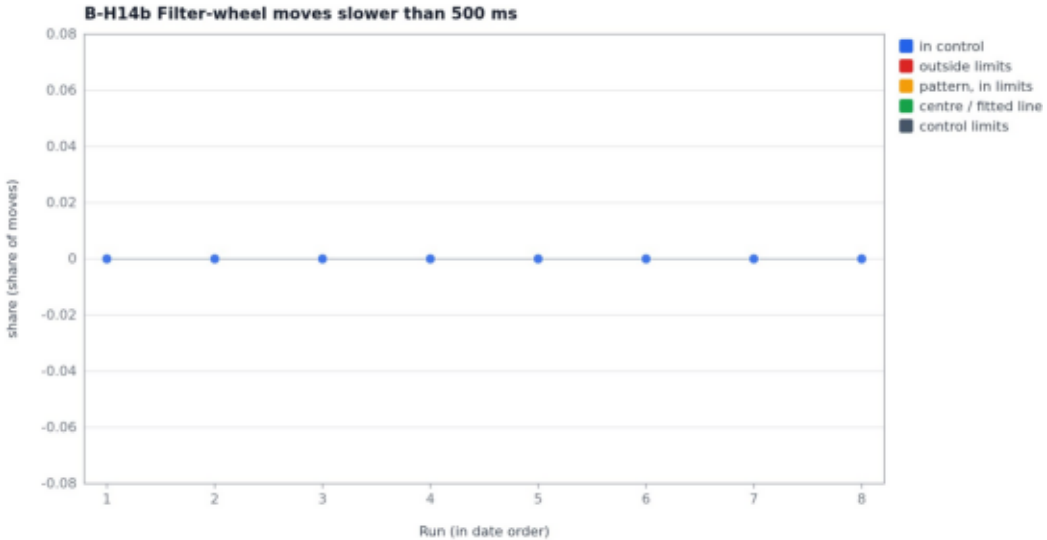
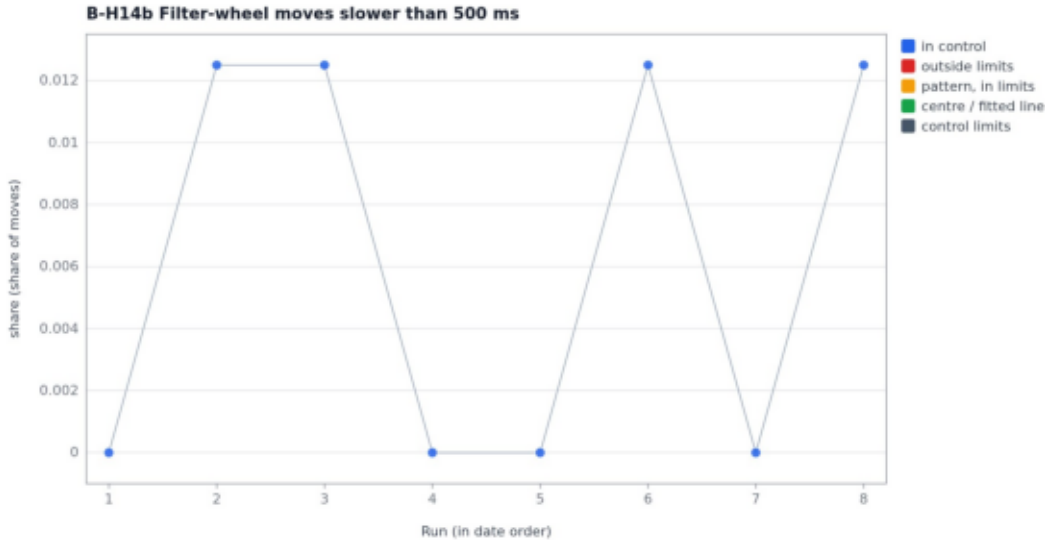
also moved by: S26 (58 %), S22 (58 %), S20 (33 %), S23 (32 %), S21 (30 %)

RUO NOT VALIDATED

cc:wheel_slow · Control charts · hardware · figure

SOUND S30_run01.eds, S30_run02.eds (+6 more)

DEVIATION S30_run19.eds, S30_run20.eds (8 files)



WHAT IT SHOWS

Occasional stalls come before a failure; any upward trend is worth a service call.

HOW TO READ THE TWO SIDES

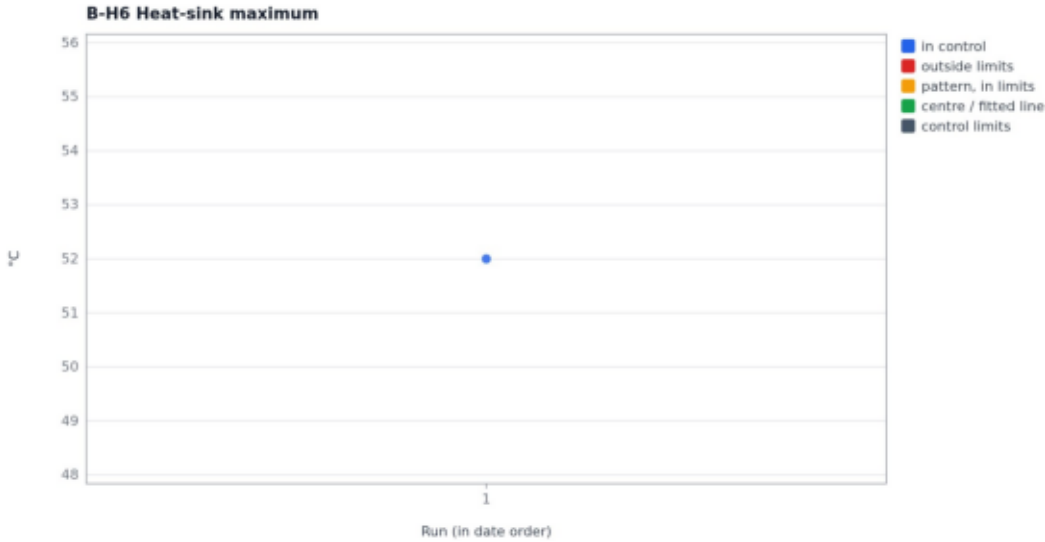
64 % of everything drawn moved between the two sides. The runs come from S30, smoothing and algorithm changed — the first eight runs of the series against its last eight. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. What matters is the level: the corpus of laboratory and vendor files puts it between 0 and 0.00313 share of moves, around 0.00313, and the generated history is written to sit inside that.

also moved by: S20 (63 %), S24 (62 %), S19 (58 %), S19 (58 %), S20 (58 %)

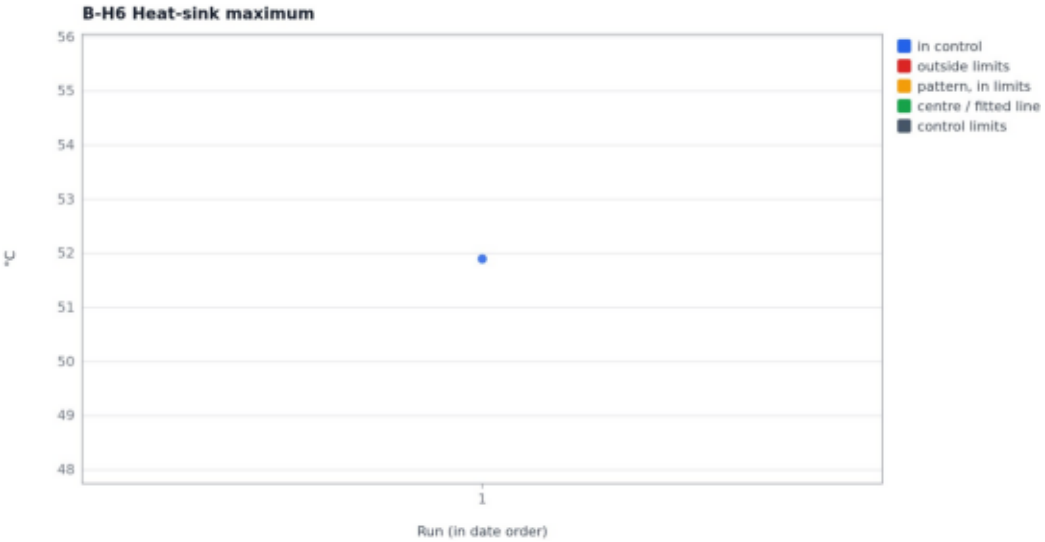
RUO NOT VALIDATED

cc:heatsink_max · Control charts · hardware · figure

SOUND S28_run01.eds



DEVIATION S28_run20.eds



WHAT IT SHOWS

Rising over time with the same protocol = cooling capacity dropping (fans, dust, room temperature).

HOW TO READ THE TWO SIDES

61 % of everything drawn moved between the two sides. The runs come from S28, calibration allowed to expire — the first run of the series against its last. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. What matters is the level: the corpus of laboratory and vendor files puts it between 50.8 and 52.6 °C, around 52, and the generated history is written to sit inside that.

also moved by: S30 (61 %), S26 (60 %), S29 (60 %), S20 (60 %), S19 (58 %)

RUO

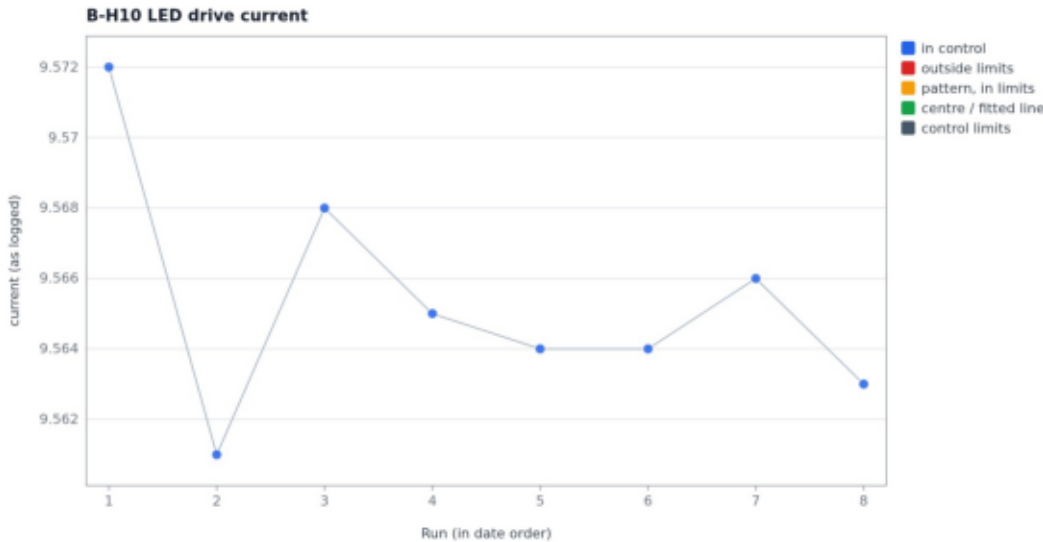
NOT VALIDATED

cc:led_current · Control charts · hardware · figure

SOUND

S25_run01.eds, S25_run02.eds

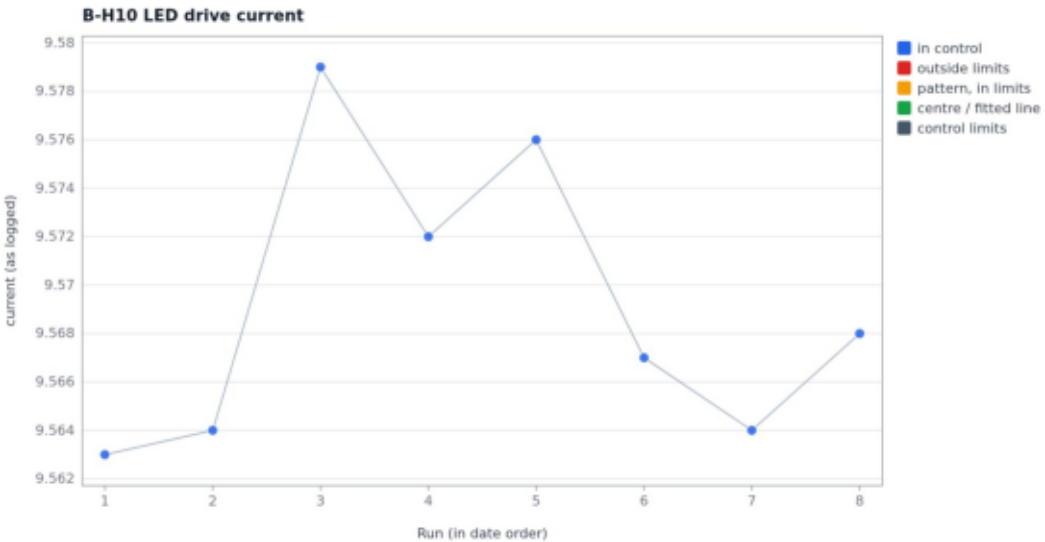
(+6 more)



DEVIATION

S25_run19.eds, S25_run20.eds

(8 files)



WHAT IT SHOWS

The instrument raises current to keep light output: a rising trend is LED ageing; the line predicts when it reaches the limit you set.

HOW TO READ THE TWO SIDES

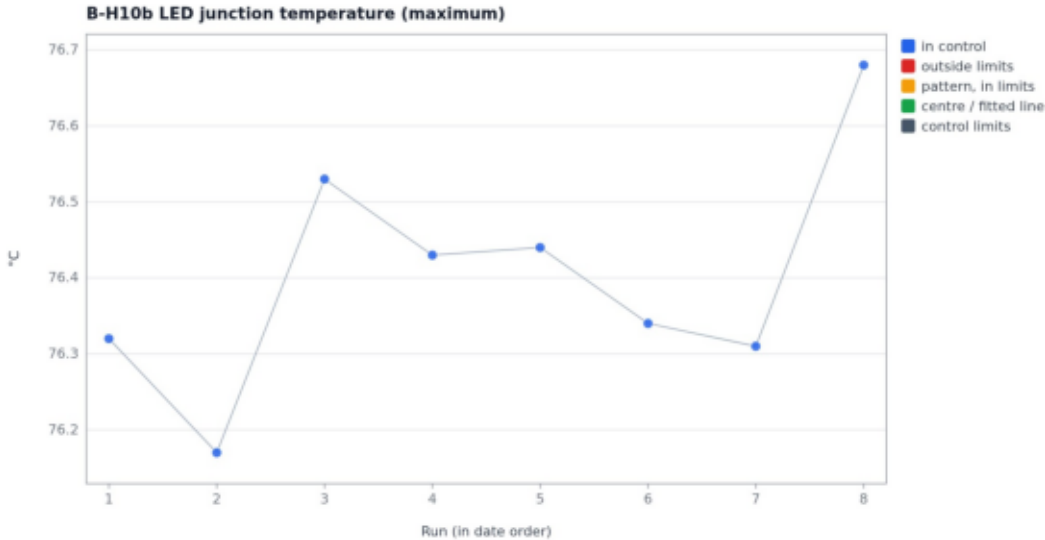
67 % of everything drawn moved between the two sides. The runs come from S25, second target of the multiplex failing — the first eight runs of the series against its last eight. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. What matters is the level: the corpus of laboratory and vendor files puts it between 9.563 and 9.574 current (as logged), around 9.574, and the generated history is written to sit inside that.

also moved by: S26 (66 %), S27 (64 %), S19 (64 %), S29 (62 %), S24 (62 %)

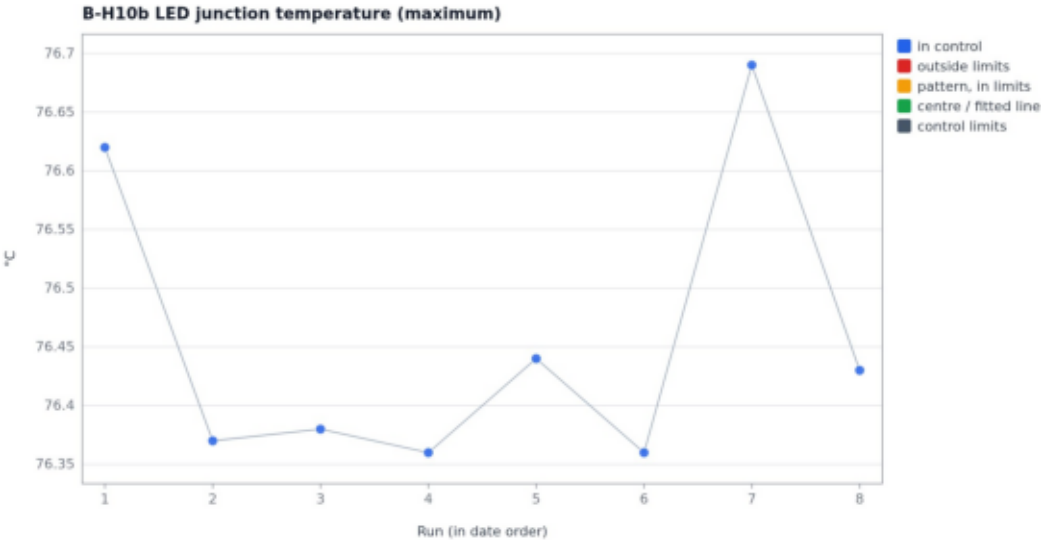
RUO NOT VALIDATED

cc:led_junction · Control charts · hardware · figure

SOUND S30_run01.eds, S30_run02.eds (+6 more)



DEVIATION S30_run19.eds, S30_run20.eds (8 files)



WHAT IT SHOWS

Higher than usual = LED cooling problem; LEDs age faster when hot.

HOW TO READ THE TWO SIDES

65 % of everything drawn moved between the two sides. The runs come from S30, smoothing and algorithm changed — the first eight runs of the series against its last eight. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. What matters is the level: the corpus of laboratory and vendor files puts it between 75.78 and 76.59 °C, around 76.57, and the generated history is written to sit inside that.

also moved by: S23 (64 %), S20 (64 %), S21 (63 %), S24 (60 %), S22 (60 %)

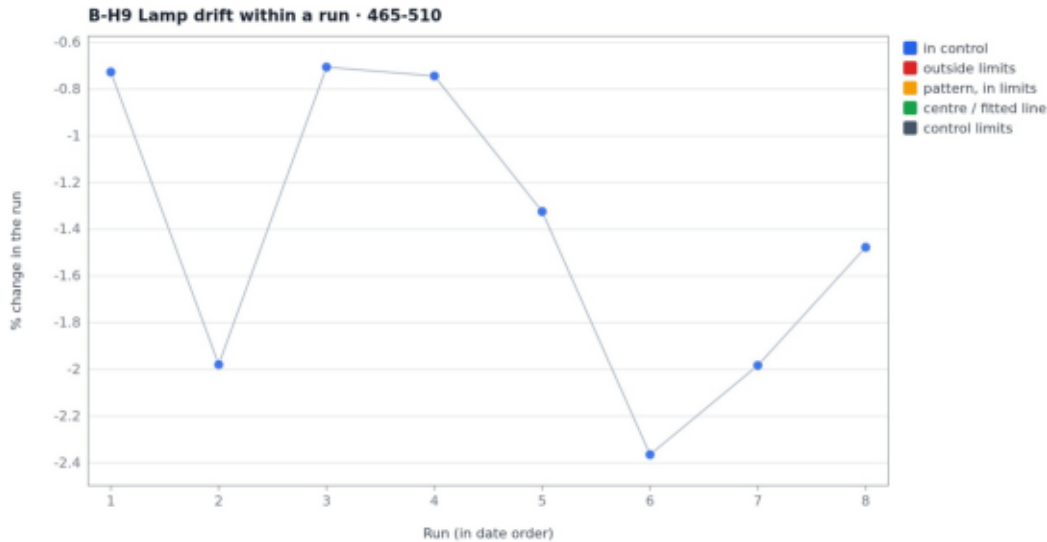
MISSY

Lamp drift within a run

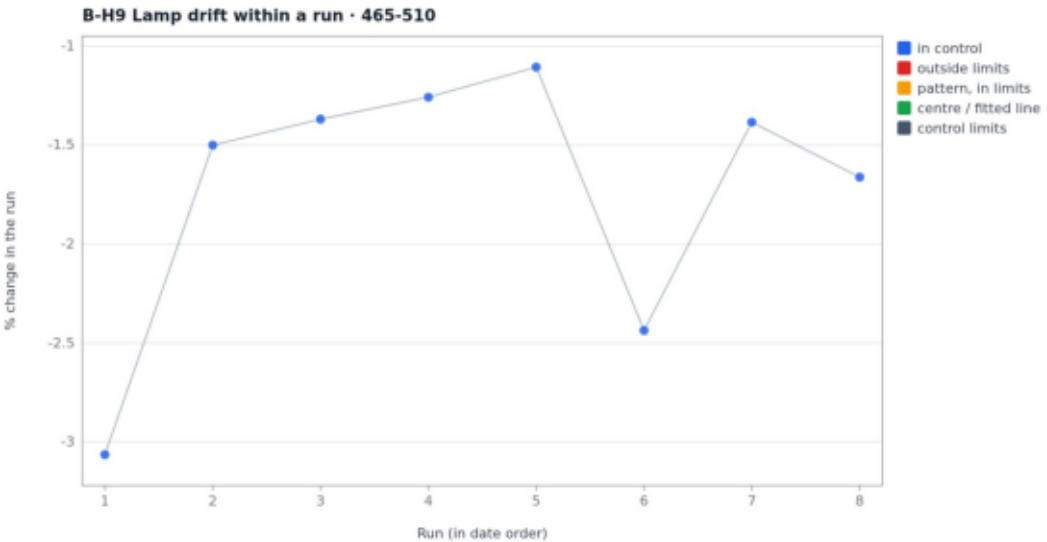
RUO NOT VALIDATED

cc:lamp_drift · Control charts · hardware · figure

SOUND S11_run01.ixo, S11_run02.ixo (+6 more)



DEVIATION S11_run19.ixo, S11_run20.ixo (8 files)



WHAT IT SHOWS

Growing drift is typical of a lamp near the end of its life.

HOW TO READ THE TWO SIDES

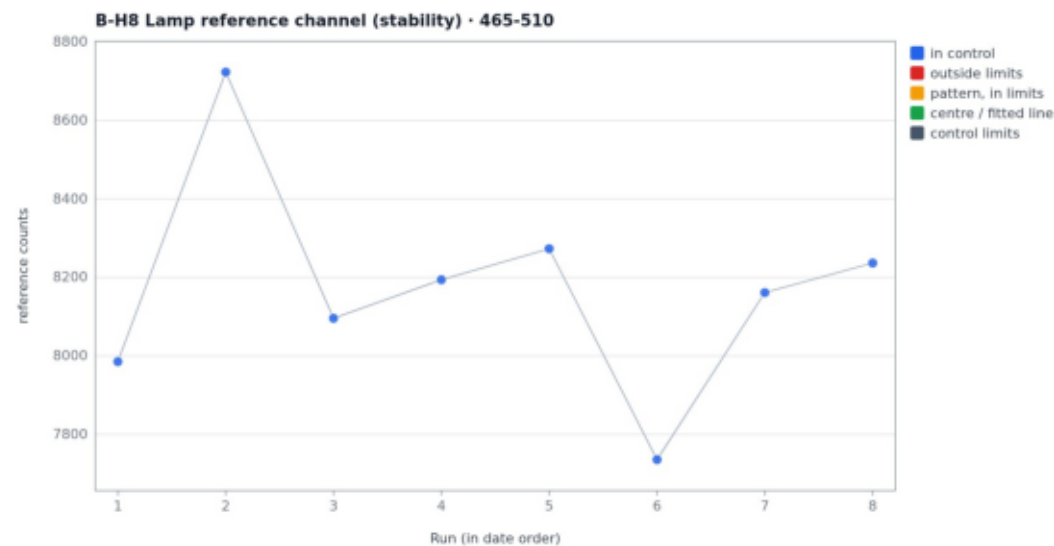
64 % of everything drawn moved between the two sides. The runs come from S11, contamination episode — the first eight runs of the series against its last eight. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. What matters is the level: the corpus of laboratory and vendor files puts it between -3.101 and -0.03301 % change in the run, around -1.456, and the generated history is written to sit inside that.

also moved by: S10 (64 %), S16 (63 %), S13 (62 %), S10 (58 %), S7 (58 %)

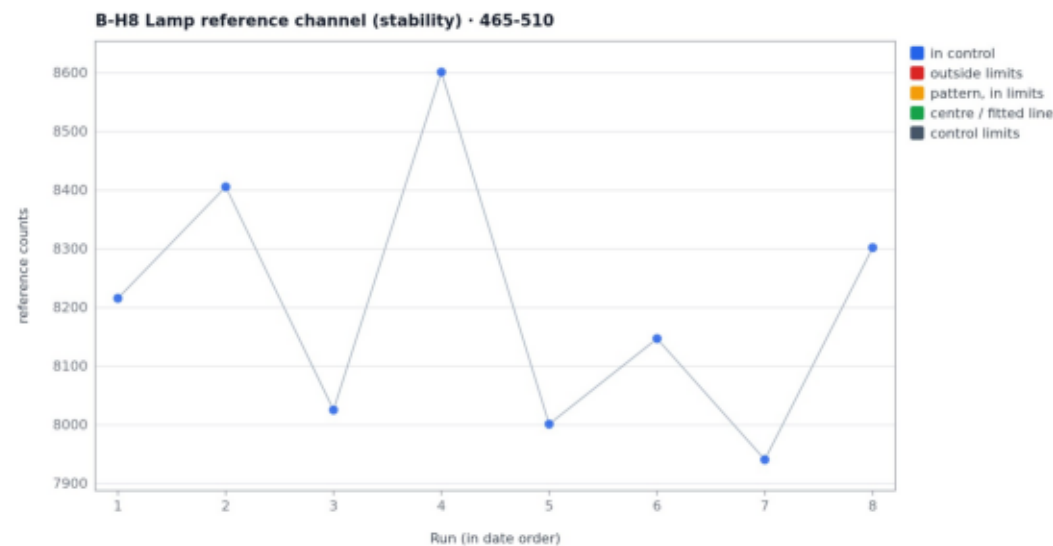
RUO NOT VALIDATED

cc:lamp_ref · Control charts · hardware · figure

SOUND S9_run01.ixo, S9_run02.ixo (+6 more)



DEVIATION S9_run19.ixo, S9_run20.ixo (8 files)



WHAT IT SHOWS

Stable = the lamp is still inside the range the instrument can compensate. A step or a point outside the limits = lamp, light guide or reference-sensor fault; check the lamp intensity % in the instrument software.

HOW TO READ THE TWO SIDES

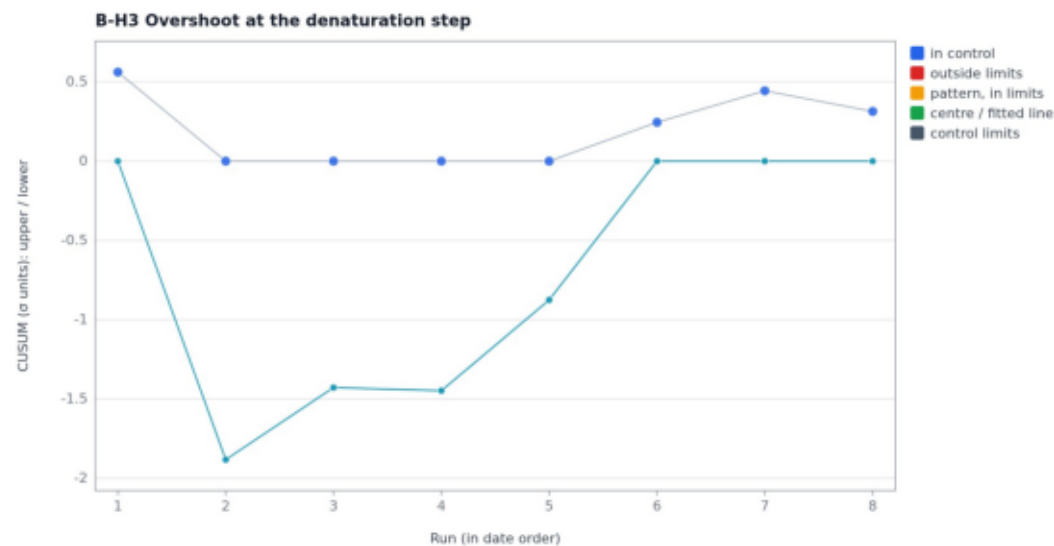
65 % of everything drawn moved between the two sides. The runs come from S9, calibrator shift between lots — the first eight runs of the series against its last eight. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. What matters is the level: the corpus of laboratory and vendor files puts it between 6282 and 9343 reference counts, around 8156, and the generated history is written to sit inside that.

also moved by: S11 (62 %), S10 (62 %), S13 (61 %), S12 (59 %), S16 (58 %)

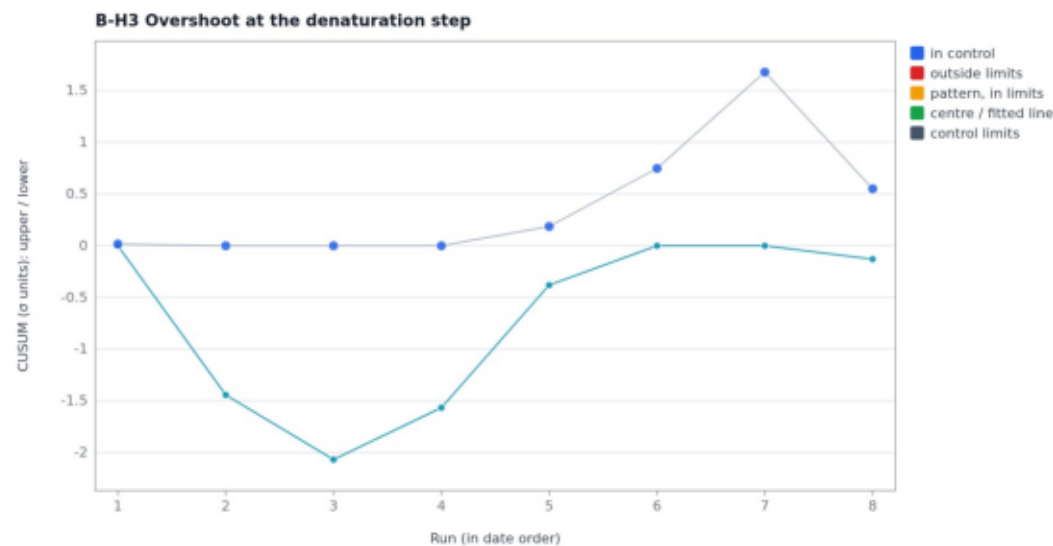
RUO NOT VALIDATED

cc:overshoot · Control charts · hardware · figure

SOUND S15_run01.ixo, S15_run02.ixo (+6 more)



DEVIATION S15_run19.ixo, S15_run20.ixo (8 files)



WHAT IT SHOWS

Watch for change, not size: a CUSUM drifting up or down means the controller or the temperature sensor has changed.

HOW TO READ THE TWO SIDES

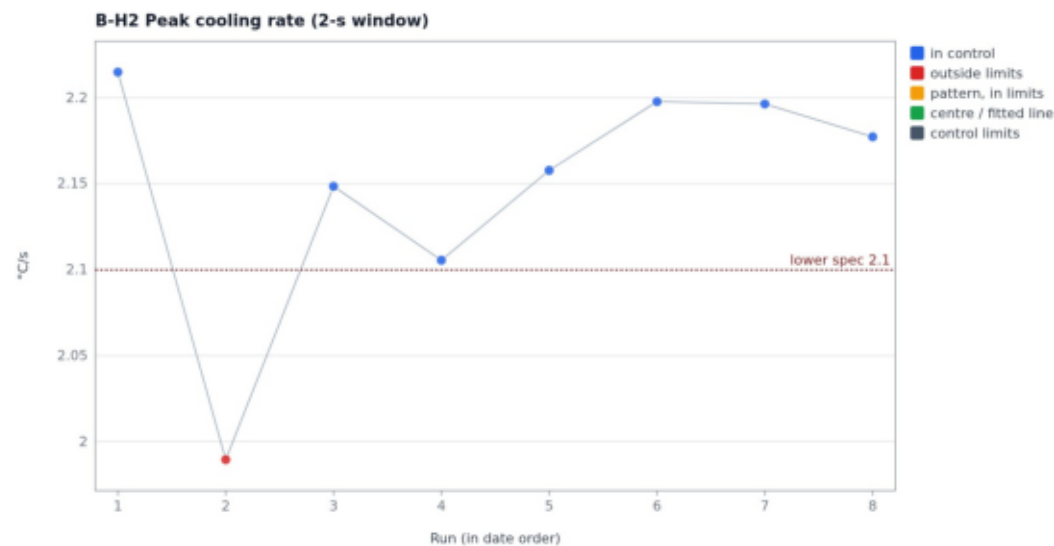
63 % of everything drawn moved between the two sides. The runs come from S15, calibration lines no longer agree — the first eight runs of the series against its last eight. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. What matters is the level: the corpus of laboratory and vendor files puts it between 2.25 and 5.47 °C, around 5.4, and the generated history is written to sit inside that.

also moved by: S14 (62 %), S12 (61 %), S11 (61 %), S16 (59 %), S13 (58 %)

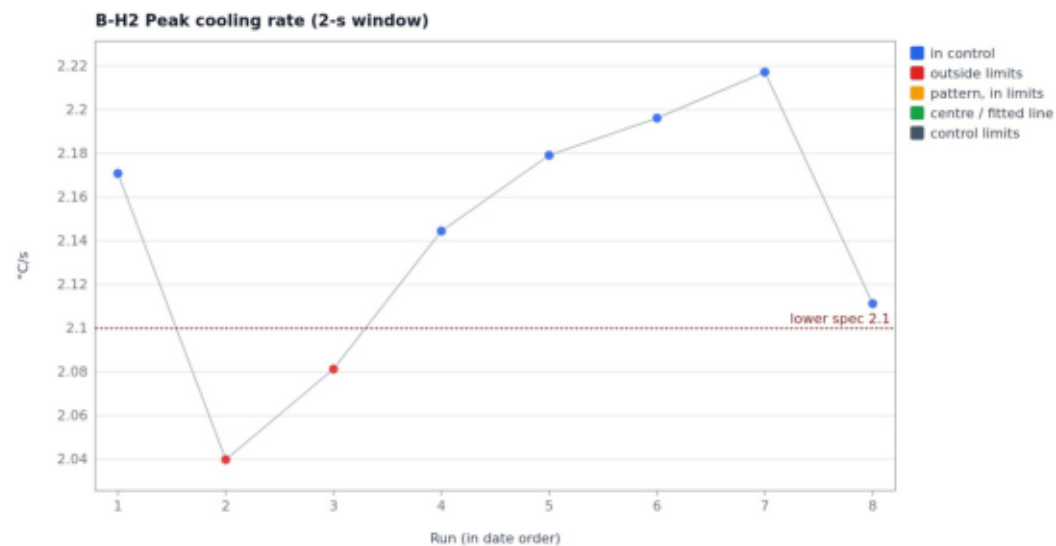
RUO NOT VALIDATED

cc:cool_rate · Control charts · hardware · figure

SOUND S15_run01.ixo, S15_run02.ixo (+6 more)



DEVIATION S15_run19.ixo, S15_run20.ixo (8 files)



WHAT IT SHOWS

Falling cooling with stable heating = clogged fan filter or failing fan (clean the filters, check fans) before Peltier replacement.

HOW TO READ THE TWO SIDES

68 % of everything drawn moved between the two sides. The runs come from S15, calibration lines no longer agree — the first eight runs of the series against its last eight. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. What matters is the level: the corpus of laboratory and vendor files puts it between 2.1 and 3.112 °C/s, around 2.448, and the generated history is written to sit inside that.

also moved by: S16 (67 %), S13 (65 %), S9 (64 %), S12 (64 %), S10 (63 %)

RUO

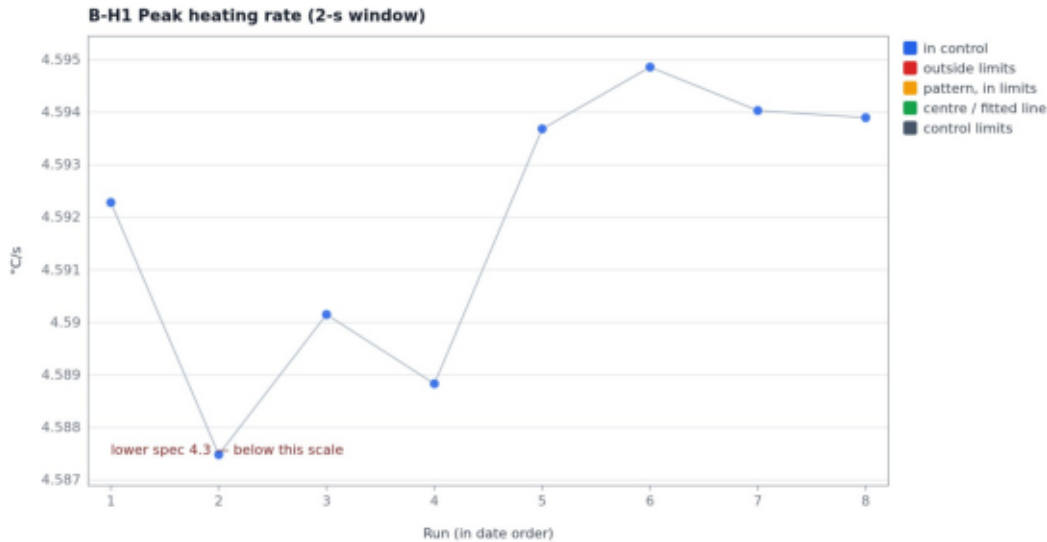
NOT VALIDATED

cc:heat_rate · Control charts · hardware · figure

SOUND

S11_run01.ixo, S11_run02.ixo

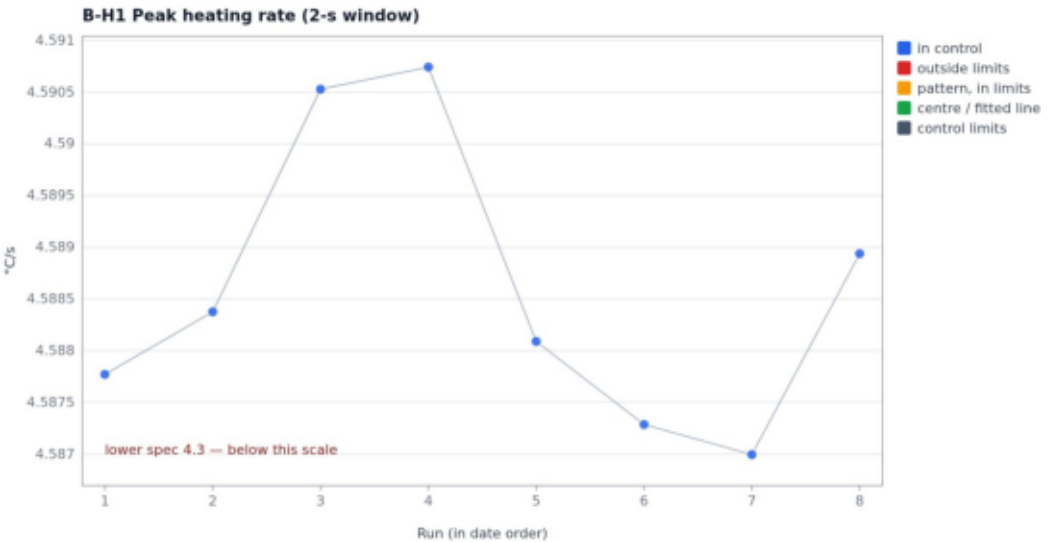
(+6 more)



DEVIATION

S11_run19.ixo, S11_run20.ixo

(8 files)



WHAT IT SHOWS

A slow fall over months is Peltier or thermal-interface ageing; the fitted line gives a conditional limit-crossing estimate, not a failure date. A single low run points to the plate or a blocked vent that day.

HOW TO READ THE TWO SIDES

66 % of everything drawn moved between the two sides. The runs come from S11, contamination episode — the first eight runs of the series against its last eight. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. What matters is the level: the corpus of laboratory and vendor files puts it between 4.3 and 5.576 °C/s, around 4.791, and the generated history is written to sit inside that.

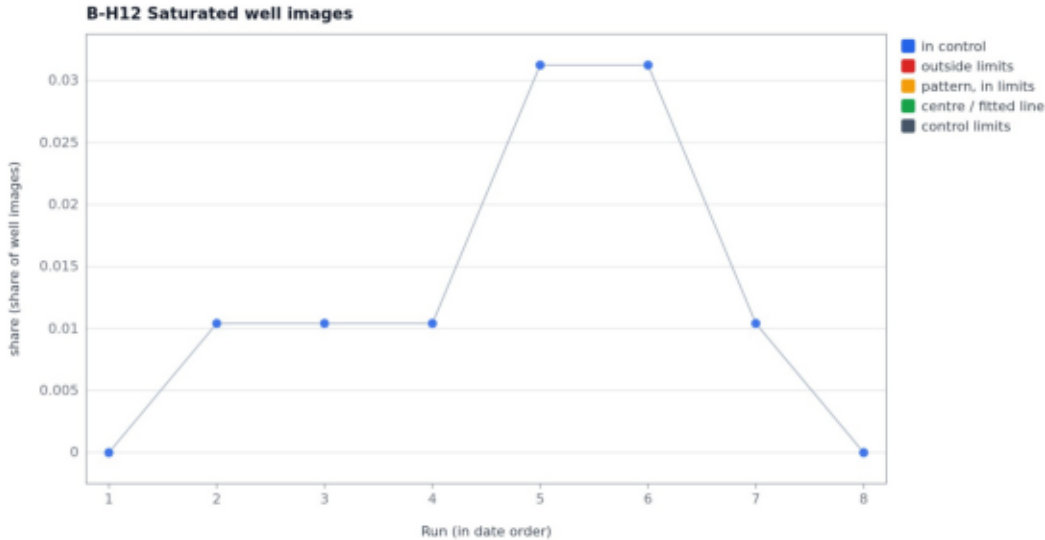
also moved by: S16 (65 %), S15 (63 %), S13 (62 %), S10 (61 %), S12 (60 %)

RUO NOT VALIDATED

cc:saturation · Control charts · hardware · figure

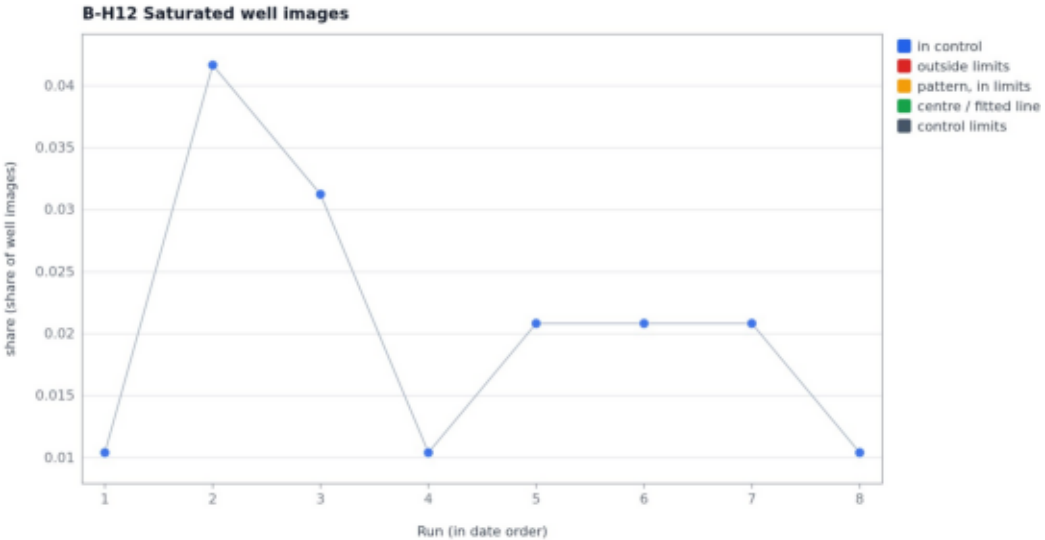
SOUND

S25_run01.eds, S25_run02.eds (+6 more)



DEVIATION

S25_run19.eds, S25_run20.eds (8 files)



WHAT IT SHOWS

A rise = exposure logic or camera fault, or overloaded samples; values above zero make the curve plateau unreliable.

HOW TO READ THE TWO SIDES

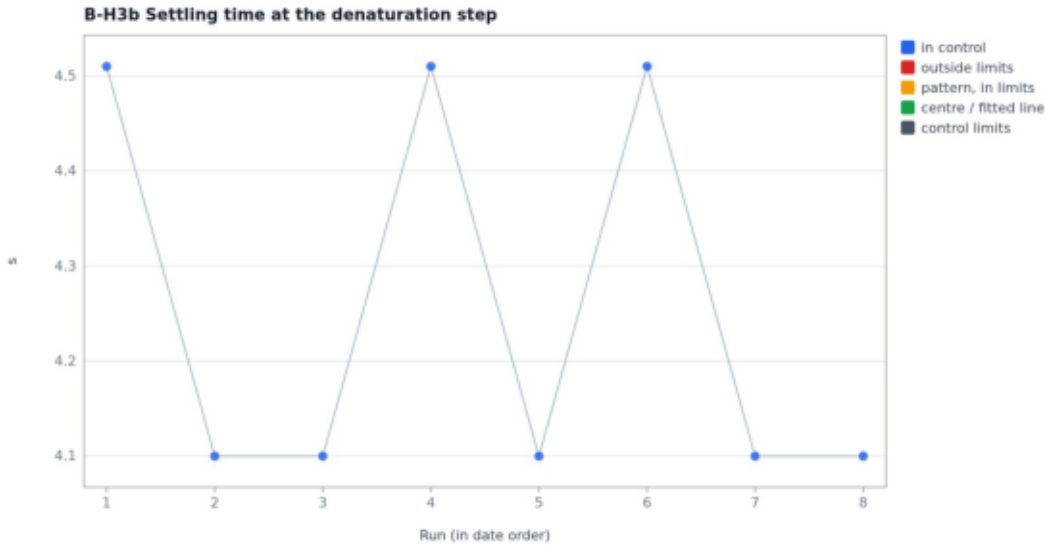
61 % of everything drawn moved between the two sides. The runs come from S25, second target of the multiplex failing — the first eight runs of the series against its last eight. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. What matters is the level: the corpus of laboratory and vendor files puts it between 0 and 0.1208 share of well images, around 0.01862, and the generated history is written to sit inside that.

also moved by: S30 (56 %), S20 (56 %), S21 (56 %), S25 (56 %), S29 (56 %)

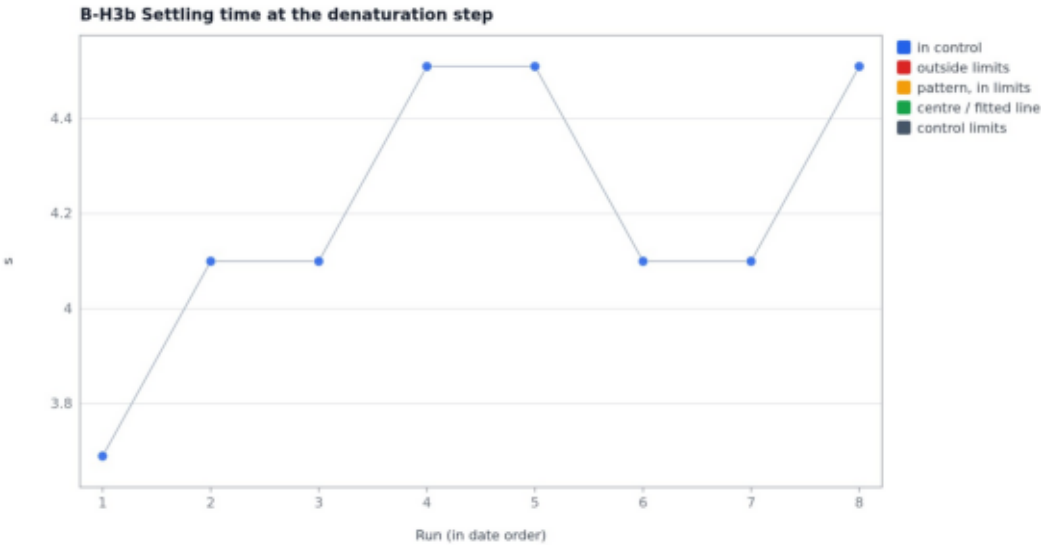
RUO NOT VALIDATED

cc:settling · Control charts · hardware · figure

SOUND S15_run01.ixx, S15_run02.ixx (+6 more)



DEVIATION S15_run19.ixx, S15_run20.ixx (8 files)



WHAT IT SHOWS

Longer settling with a normal ramp rate points to control tuning or sensor lag.

HOW TO READ THE TWO SIDES

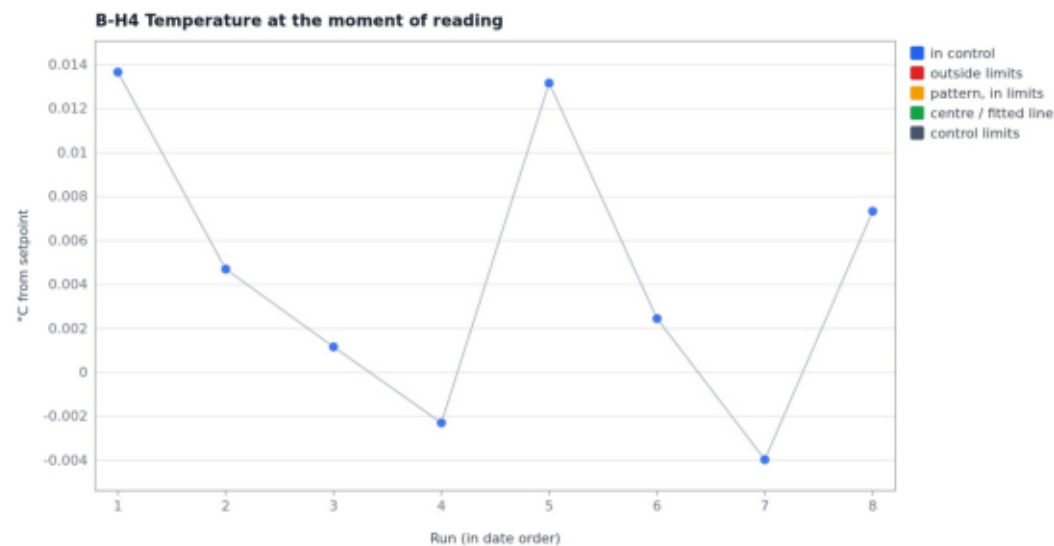
58 % of everything drawn moved between the two sides. The runs come from S15, calibration lines no longer agree — the first eight runs of the series against its last eight. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. What matters is the level: the corpus of laboratory and vendor files puts it between 3.816 and 4.617 s, around 4.533, and the generated history is written to sit inside that.

also moved by: S13 (52 %), S2 (45 %), S4 (45 %), S10 (45 %), S11 (45 %)

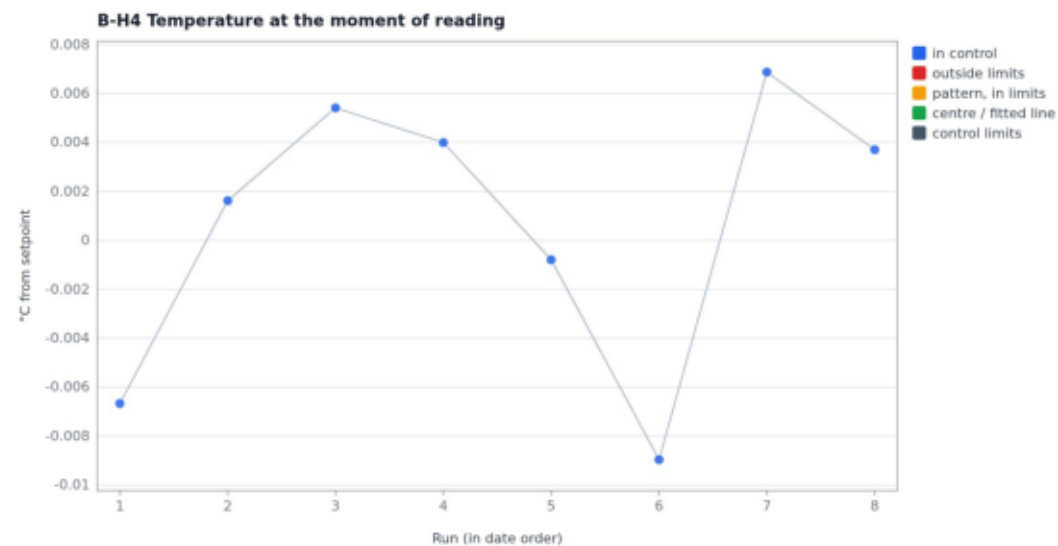
RUO NOT VALIDATED

cc:hold_temp · Control charts · hardware · figure

SOUND S27_run01.eds, S27_run02.eds (+6 more)



DEVIATION S27_run19.eds, S27_run20.eds (8 files)



WHAT IT SHOWS

X drifting away from its usual offset = temperature calibration drift (verify with a temperature plate). S rising = unstable control.

HOW TO READ THE TWO SIDES

66 % of everything drawn moved between the two sides. The runs come from S27, reagent lot change — the first eight runs of the series against its last eight. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. No level for this chart could be measured in the corpus, so the generated history carries the theoretical range instead.

also moved by: S28 (64 %), S24 (64 %), S26 (63 %), S30 (62 %), S19 (61 %)

RUO

NOT VALIDATED

cc:transition

·

Control charts

·

hardware

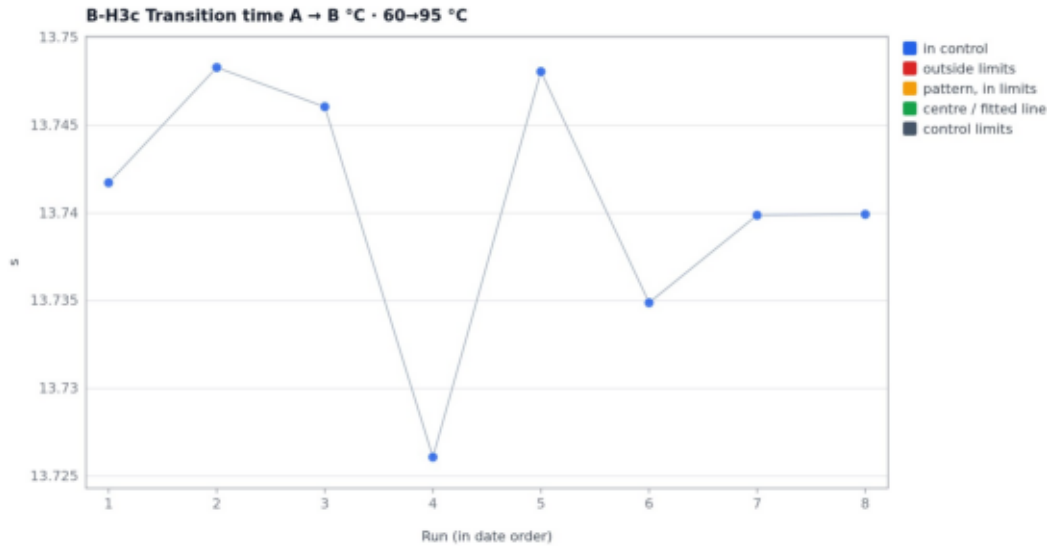
·

figure

SOUND

S21_run01.eds, S21_run02.eds

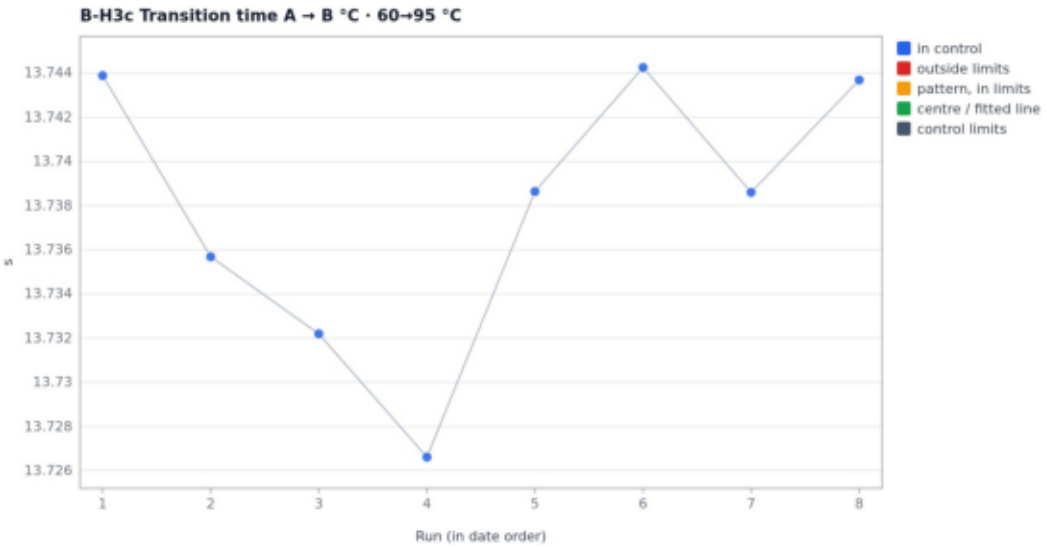
(+6 more)



DEVIATION

S21_run19.eds, S21_run20.eds

(8 files)



WHAT IT SHOWS

Compare a transition only with itself and the same program. A cooling transition that lengthens while heating transitions stay flat points to heat removal (fans, filters, heat sink); all transitions lengthening together points to the Peltier elements or the room.

HOW TO READ THE TWO SIDES

66 % of everything drawn moved between the two sides. The runs come from S21, threshold switched from automatic to manual — the first eight runs of the series against its last eight. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. What matters is the level: the corpus of laboratory and vendor files puts it between 13.72 and 14.65 s, around 13.75, and the generated history is written to sit inside that.

also moved by: S25 (66 %), S13 (65 %), S12 (65 %), S20 (65 %), S10 (64 %)

RUO NOT VALIDATED

cc:efficiency · Control charts · method · figure

SOUND S4_normal.ix



DEVIATION S4_abnormal.ix

No data for this chart in the loaded runs or the imported history.

WHAT IT SHOWS

A slow drop = degrading standard or reagent; outside 90–110 % fails MIQE-style acceptance.

HOW TO READ THE TWO SIDES

100 % of everything drawn moved between the two sides. The runs come from S4, wrong dilution — the sound file against the deviating one. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. No level for this chart could be measured in the corpus, so the generated history carries the theoretical range instead.

also moved by: S12 (66 %), S28 (63 %), S15 (63 %), S22 (62 %), S10 (62 %)

RUO

NOT VALIDATED

cc:rox_level

·

Control charts

·

method

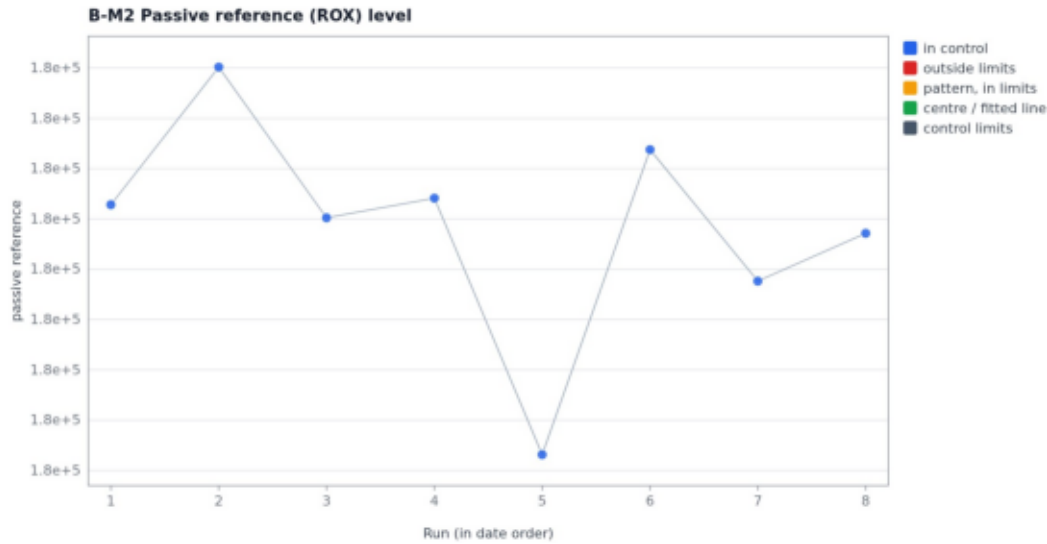
·

figure

SOUND

S22_run01.eds, S22_run02.eds

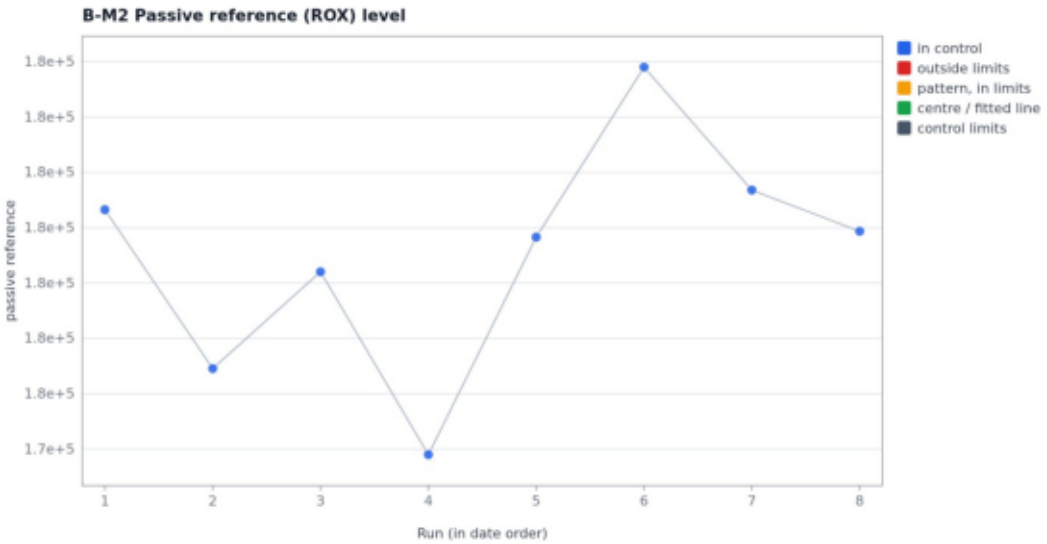
(+6 more)



DEVIATION

S22_run19.eds, S22_run20.eds

(8 files)



WHAT IT SHOWS

A step = new master-mix lot or ROX concentration.

HOW TO READ THE TWO SIDES

66 % of everything drawn moved between the two sides. The runs come from S22, failing wells omitted instead of repeated — the first eight runs of the series against its last eight. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. What matters is the level: the corpus of laboratory and vendor files puts it between 1.468e+05 and 2.879e+05 passive reference, around 1.781e+05, and the generated history is written to sit inside that.

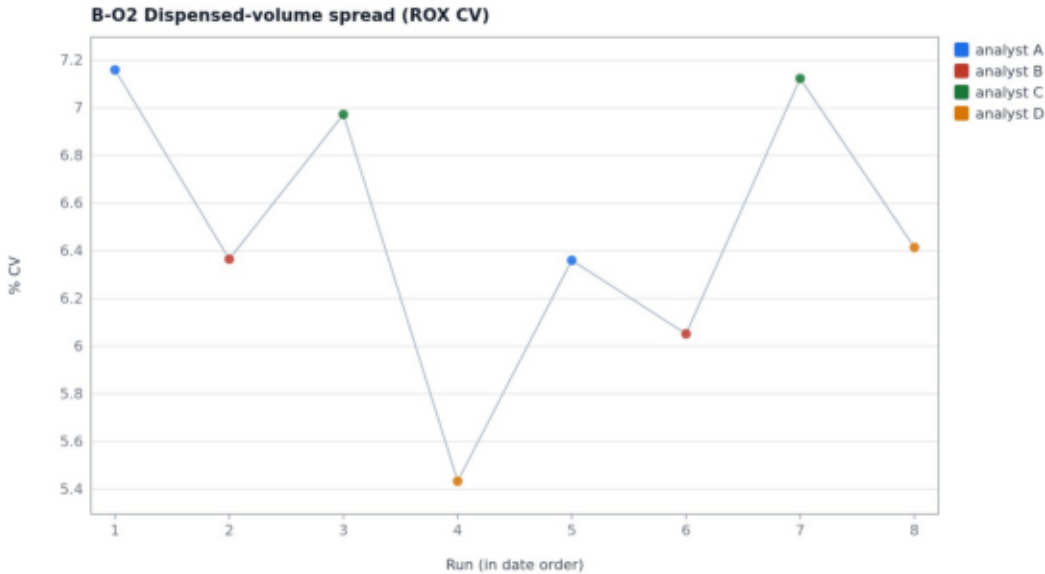
also moved by: S28 (65 %), S27 (64 %), S25 (62 %), S20 (62 %), S29 (61 %)

RUO NOT VALIDATED

cc:rox_cv · Control charts · operator · figure

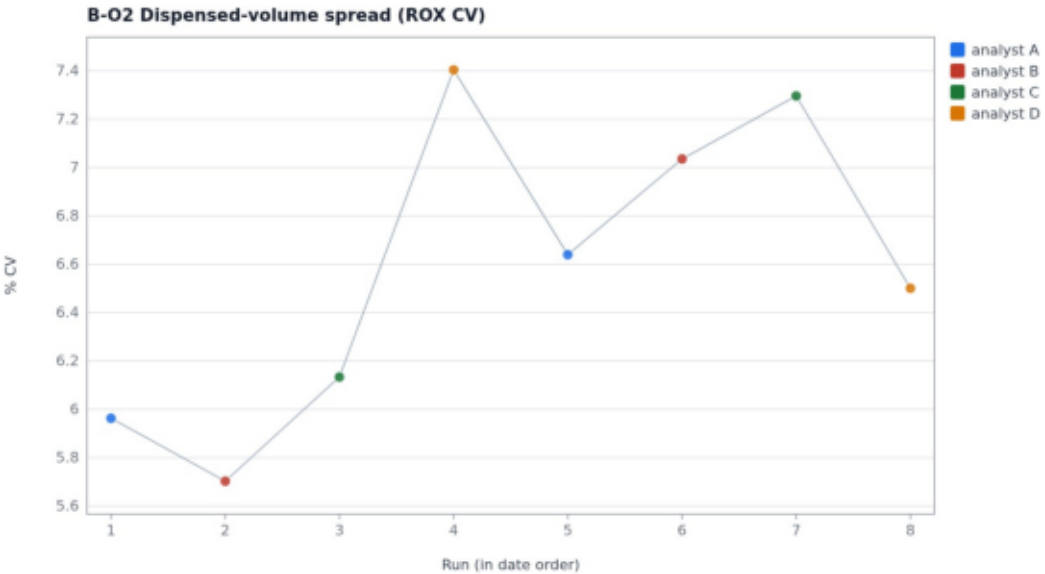
SOUND

S20_run01.eds, S20_run02.eds (+6 more)



DEVIATION

S20_run19.eds, S20_run20.eds (8 files)



WHAT IT SHOWS

High CV = uneven pipetting or bubbles.

HOW TO READ THE TWO SIDES

74 % of everything drawn moved between the two sides. The runs come from S20, baseline window changed by hand — the first eight runs of the series against its last eight. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. What matters is the level: the corpus of laboratory and vendor files puts it between 6.339 and 15.76 % CV, around 6.371, and the generated history is written to sit inside that.

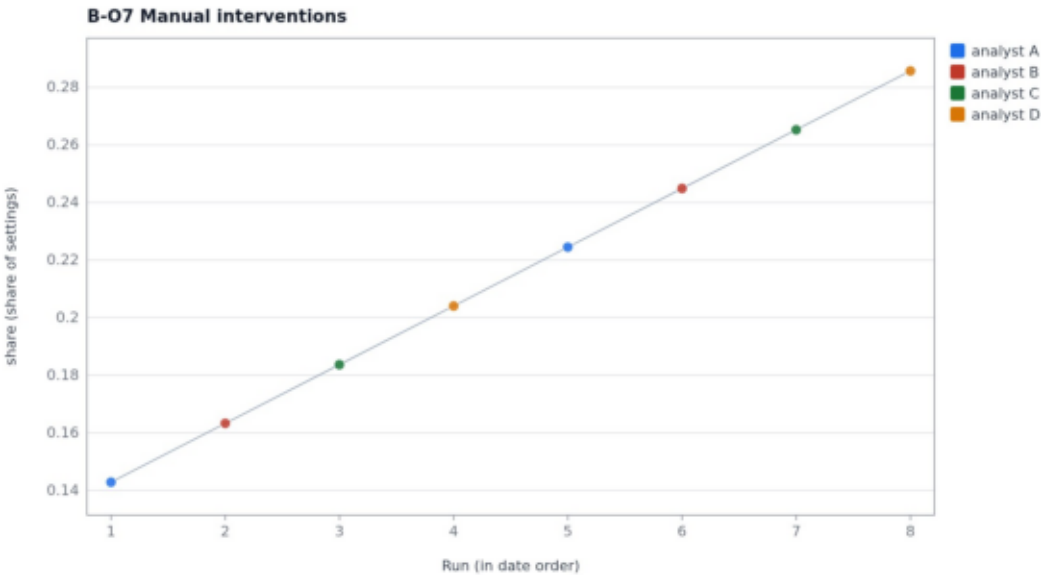
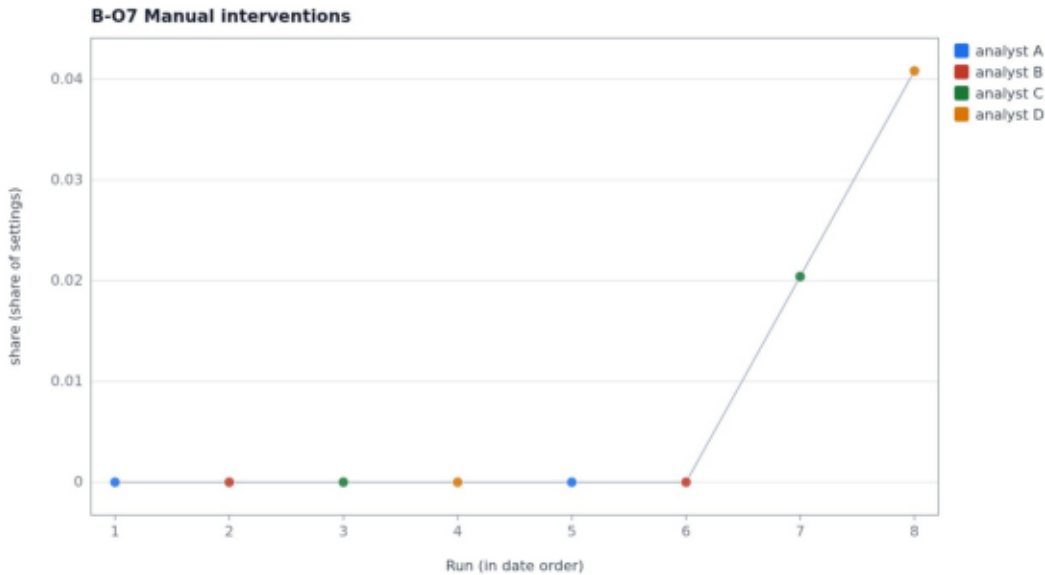
also moved by: S29 (66 %), S21 (66 %), S28 (65 %), S23 (65 %), S25 (65 %)

RUO NOT VALIDATED

cc:manual_rate · Control charts · operator · figure

SOUND S22_run01.eds, S22_run02.eds (+6 more)

DEVIATION S22_run19.eds, S22_run20.eds (8 files)



WHAT IT SHOWS

High or rising → ask for documented reasons (OIG report lessons); compare operators with the funnel view.

HOW TO READ THE TWO SIDES

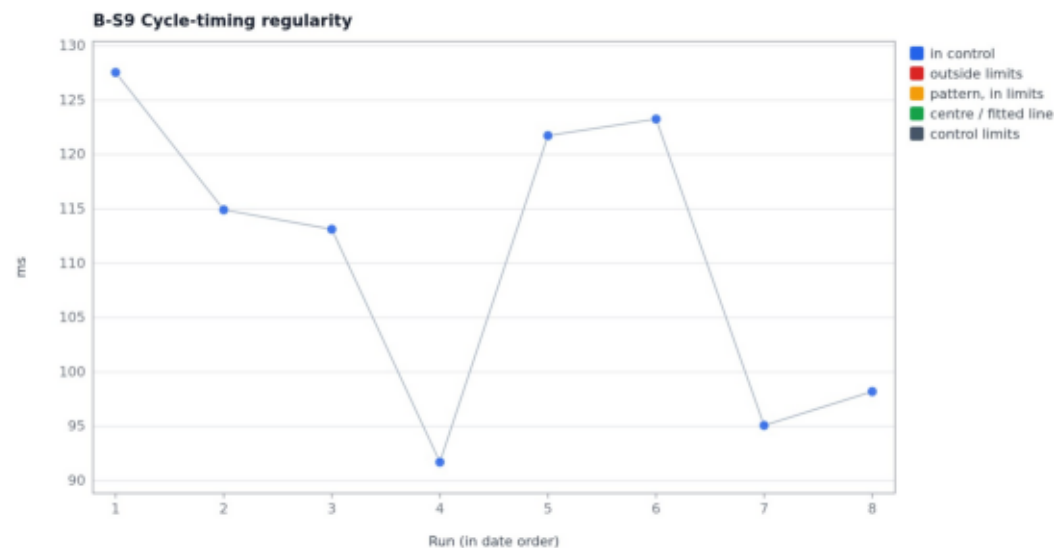
61 % of everything drawn moved between the two sides. The runs come from S22, failing wells omitted instead of repeated — the first eight runs of the series against its last eight. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. What matters is the level: the corpus of laboratory and vendor files puts it between 0 and 0 share of settings, around 0, and the generated history is written to sit inside that.

also moved by: S20 (52 %), S21 (52 %), S20 (51 %), S21 (51 %), S22 (49 %)

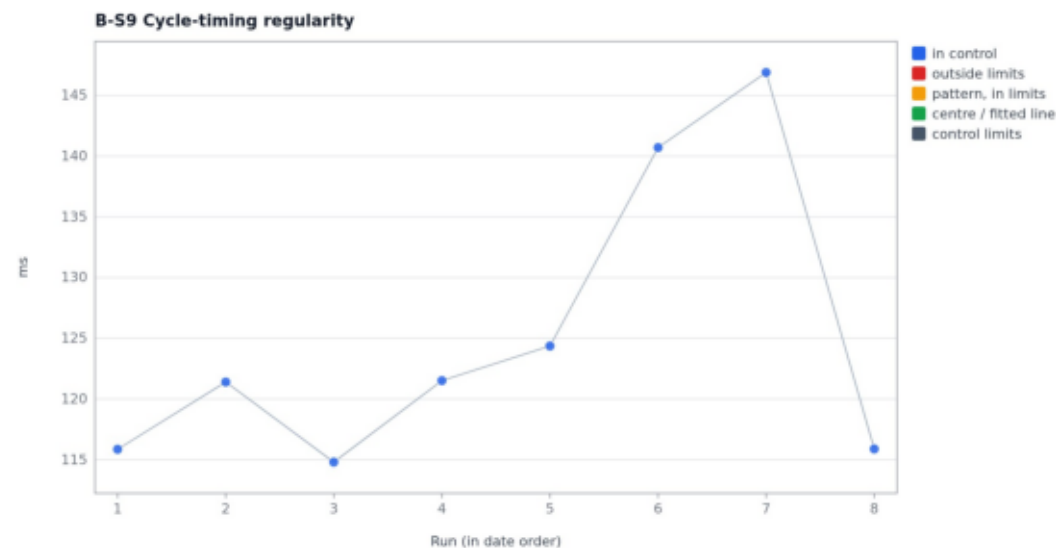
RUO NOT VALIDATED

cc:cycle_sd · Control charts · software · figure

SOUND S26_run01.eds, S26_run02.eds (+6 more)



DEVIATION S26_run19.eds, S26_run20.eds (8 files)



WHAT IT SHOWS

Larger = scheduling problems in the instrument.

HOW TO READ THE TWO SIDES

66 % of everything drawn moved between the two sides. The runs come from S26, relative-quantification reference changed — the first eight runs of the series against its last eight. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. What matters is the level: the corpus of laboratory and vendor files puts it between 99.83 and 146.8 ms, around 108.4, and the generated history is written to sit inside that.

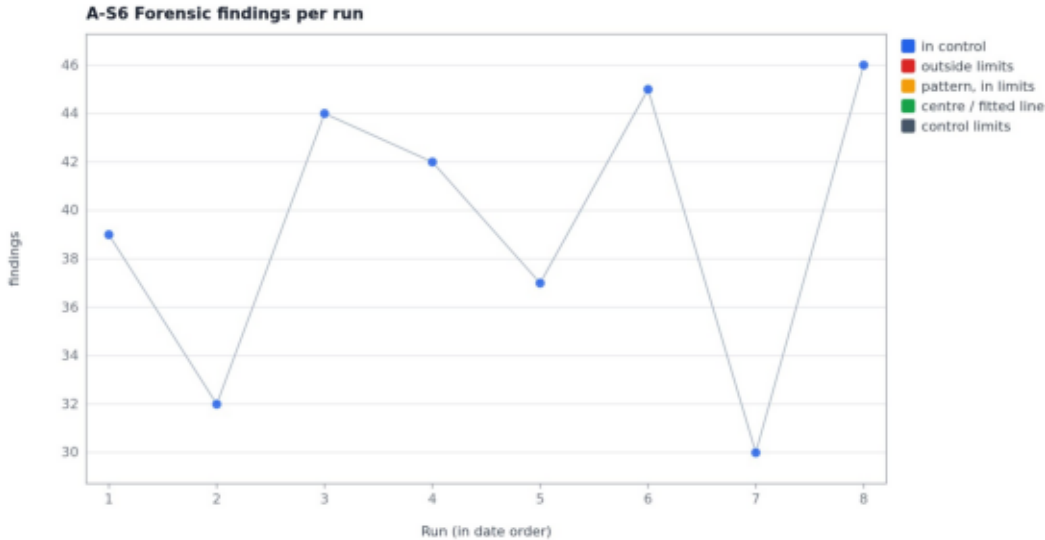
also moved by: S13 (66 %), S15 (66 %), S16 (65 %), S19 (63 %), S23 (63 %)

RUO NOT VALIDATED

cc:findings · Control charts · software · figure

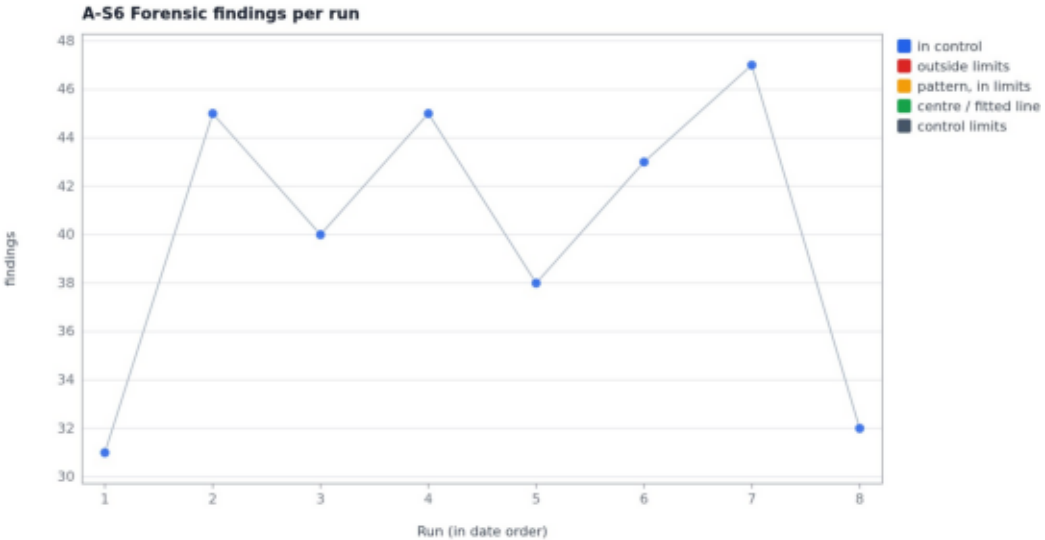
SOUND

S22_run01.eds, S22_run02.eds (+6 more)



DEVIATION

S22_run19.eds, S22_run20.eds (8 files)



WHAT IT SHOWS

A run above the limit needs a closer look in the forensic log.

HOW TO READ THE TWO SIDES

68 % of everything drawn moved between the two sides. The runs come from S22, failing wells omitted instead of repeated — the first eight runs of the series against its last eight. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. What matters is the level: the corpus of laboratory and vendor files puts it between 1 and 7 findings, around 1, and the generated history is written to sit inside that.

also moved by: S43 (63 %), S28 (63 %), S23 (63 %), S19 (63 %), S36 (63 %)

RUO

NOT VALIDATED

cc:tick_late

·

Control charts

·

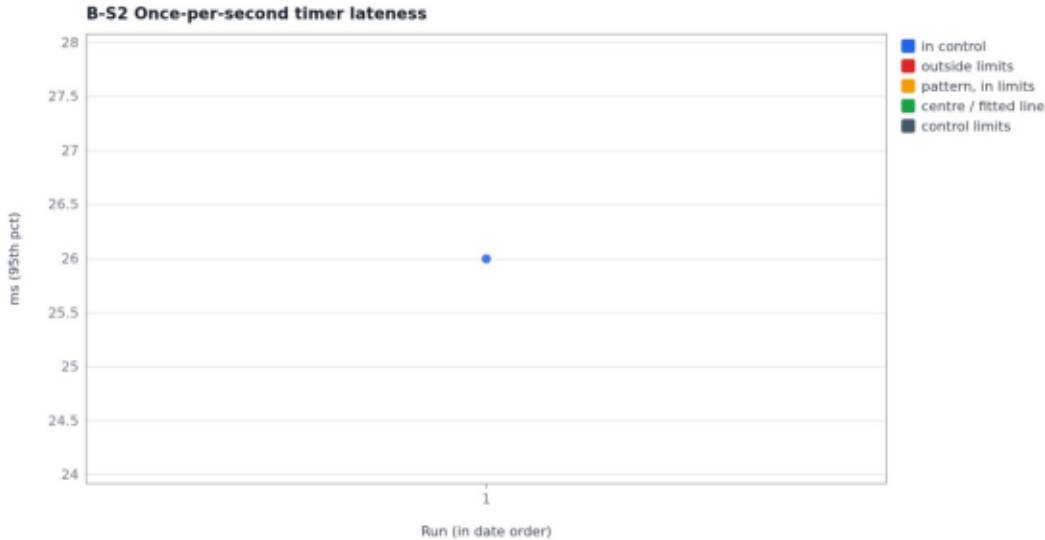
software

·

figure

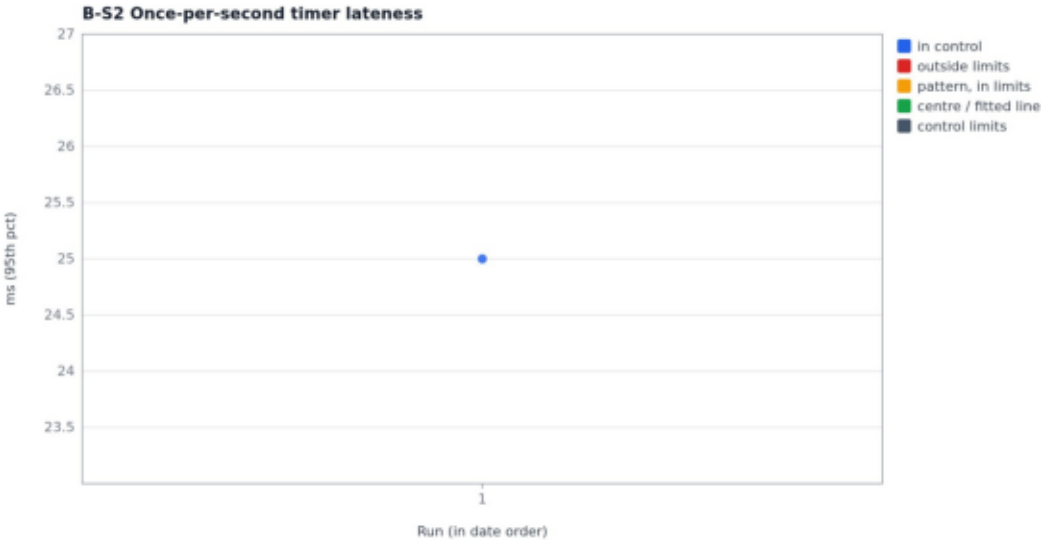
SOUND

S21_run01.eds



DEVIATION

S21_run20.eds



WHAT IT SHOWS

Rising = the embedded CPU is saturated (mostly while imaging).

HOW TO READ THE TWO SIDES

56 % of everything drawn moved between the two sides. The runs come from S21, threshold switched from automatic to manual — the first run of the series against its last. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. What matters is the level: the corpus of laboratory and vendor files puts it between 10 and 68 ms (95th pct), around 26, and the generated history is written to sit inside that.

also moved by: S23 (56 %), S19 (55 %), S23 (55 %), S20 (55 %), S24 (55 %)

RUO

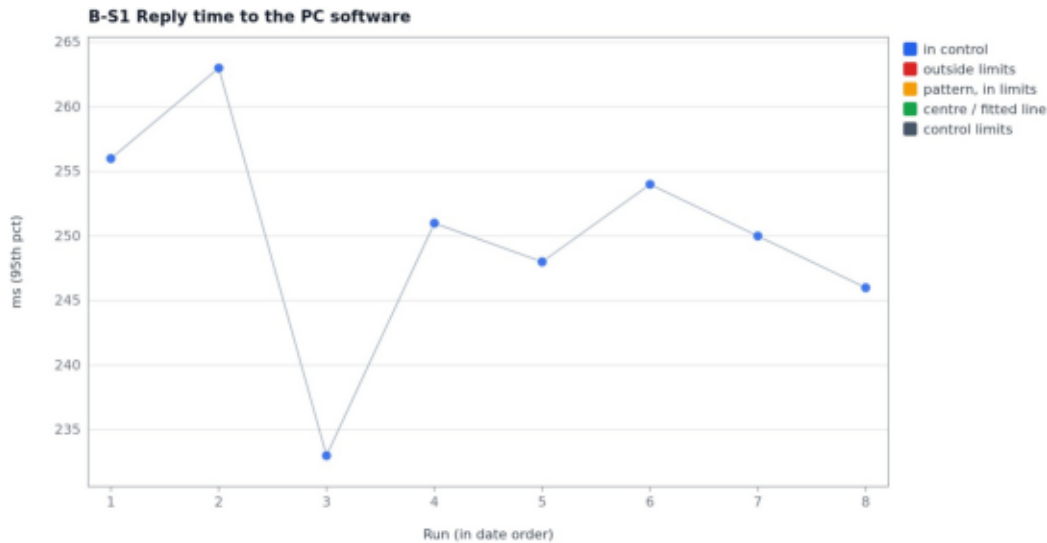
NOT VALIDATED

cc:cmd_rt · Control charts · software · figure

SOUND

S27_run01.eds, S27_run02.eds

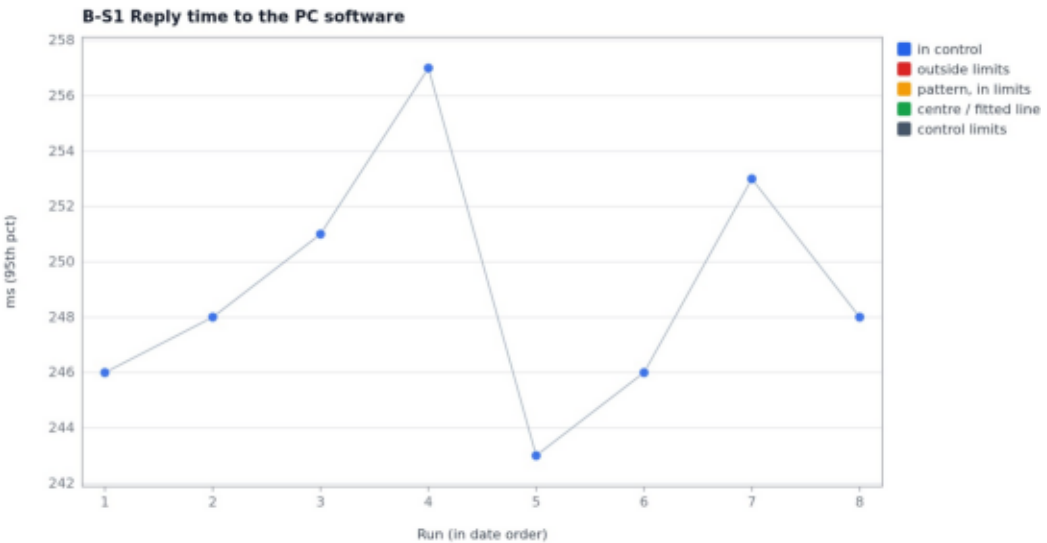
(+6 more)



DEVIATION

S27_run19.eds, S27_run20.eds

(8 files)



WHAT IT SHOWS

Rising = controller overloaded or network problem.

HOW TO READ THE TWO SIDES

66 % of everything drawn moved between the two sides. The runs come from S27, reagent lot change — the first eight runs of the series against its last eight. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. What matters is the level: the corpus of laboratory and vendor files puts it between 159 and 294 ms (95th pct), around 198, and the generated history is written to sit inside that.

also moved by: S26 (63 %), S29 (63 %), S21 (61 %), S28 (60 %), S23 (60 %)

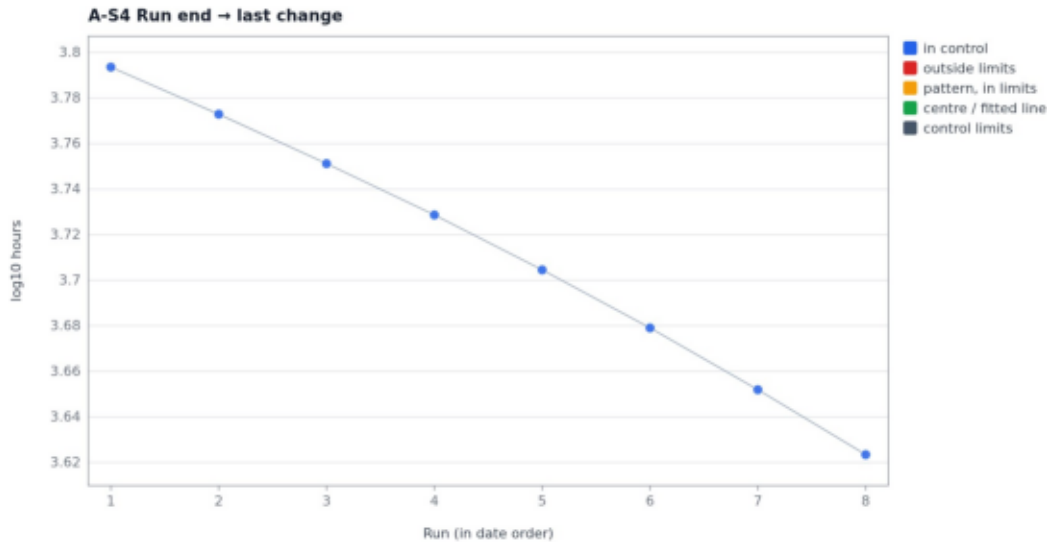
MISSY

Run end → last change

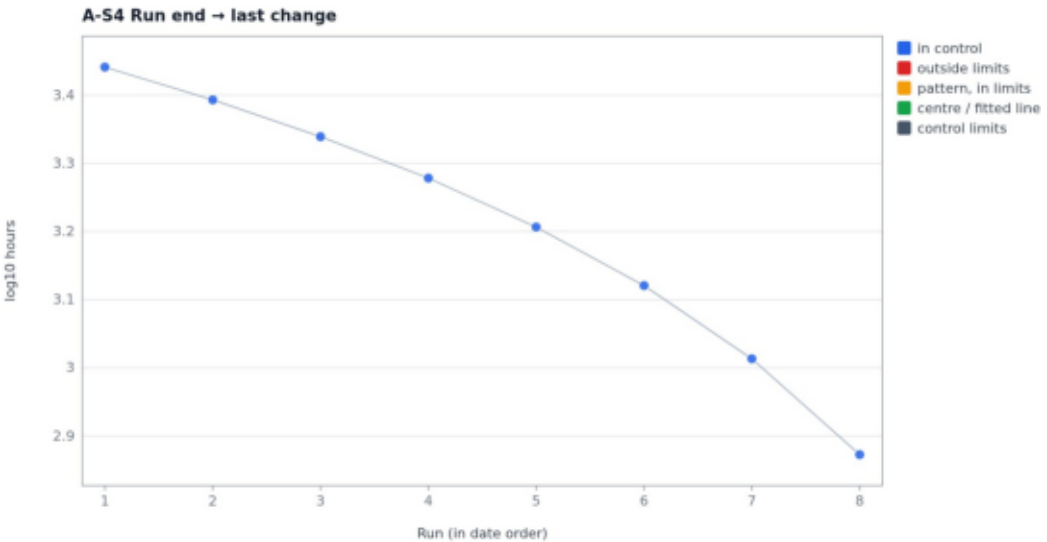
RUO NOT VALIDATED

cc:edit_delay · Control charts · software · figure

SOUND S19_run01.eds, S19_run02.eds (+6 more)



DEVIATION S19_run19.eds, S19_run20.eds (8 files)



WHAT IT SHOWS

Points above the limit are late edits; cross-check with the operator group.

HOW TO READ THE TWO SIDES

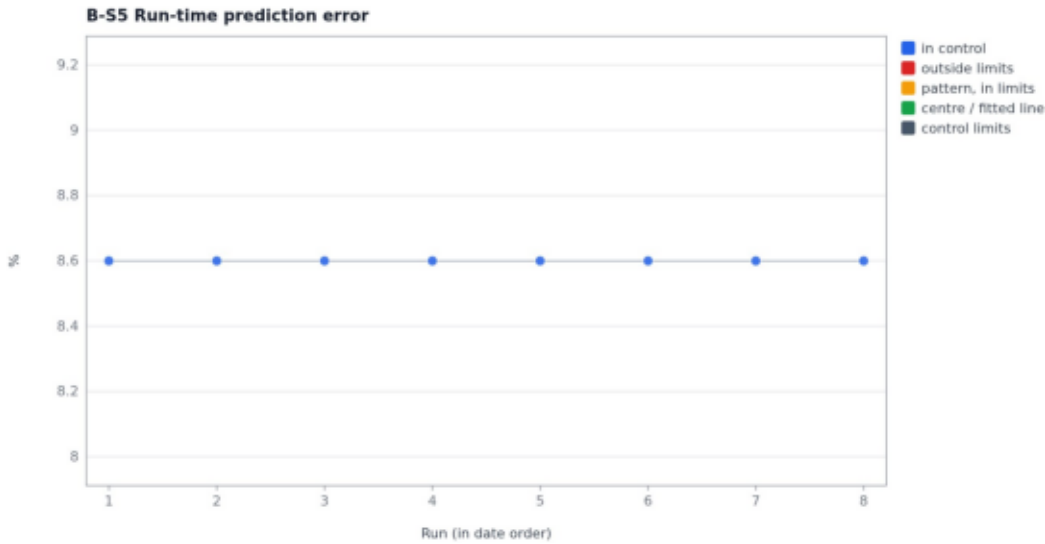
65 % of everything drawn moved between the two sides. The runs come from S19, passive reference falling away — the first eight runs of the series against its last eight. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. No level for this chart could be measured in the corpus, so the generated history carries the theoretical range instead.

also moved by: S20 (65 %), S21 (65 %), S22 (65 %), S23 (65 %), S24 (65 %)

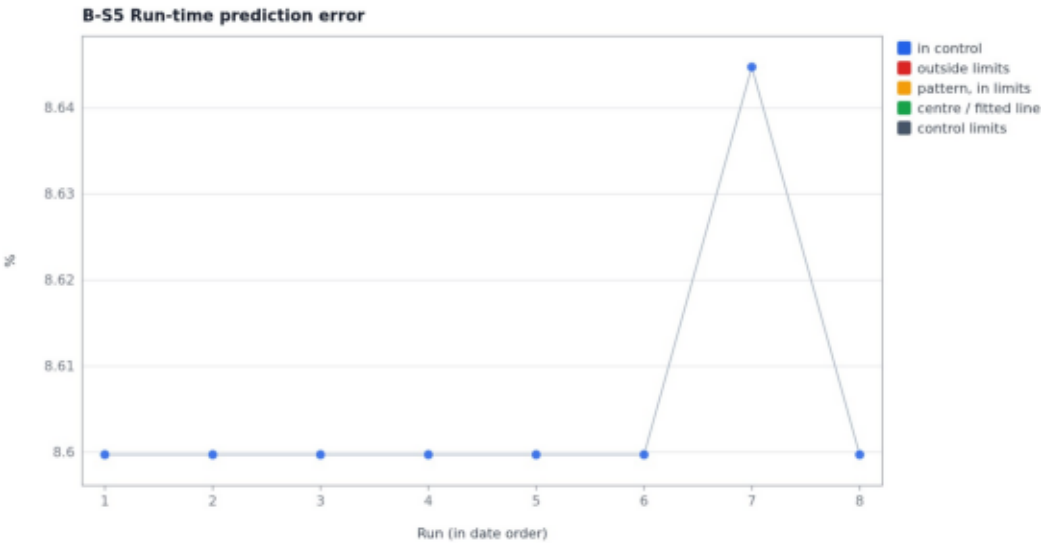
RUO NOT VALIDATED

cc:runtime_err · Control charts · software · figure

SOUND S29_run01.eds, S29_run02.eds (+6 more)



DEVIATION S29_run19.eds, S29_run20.eds (8 files)



WHAT IT SHOWS

Creeping up = actions take longer than the firmware expects.

HOW TO READ THE TWO SIDES

59 % of everything drawn moved between the two sides. The runs come from S29, block and instrument swapped — the first eight runs of the series against its last eight. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. What matters is the level: the corpus of laboratory and vendor files puts it between 8.373 and 11.41 %, around 8.534, and the generated history is written to sit inside that.

also moved by: S1 (0 %), S2 (0 %), S3 (0 %), S4 (0 %), S5 (0 %)

RUO

NOT VALIDATED

cc:ct_mismatch

·

Control charts

·

software

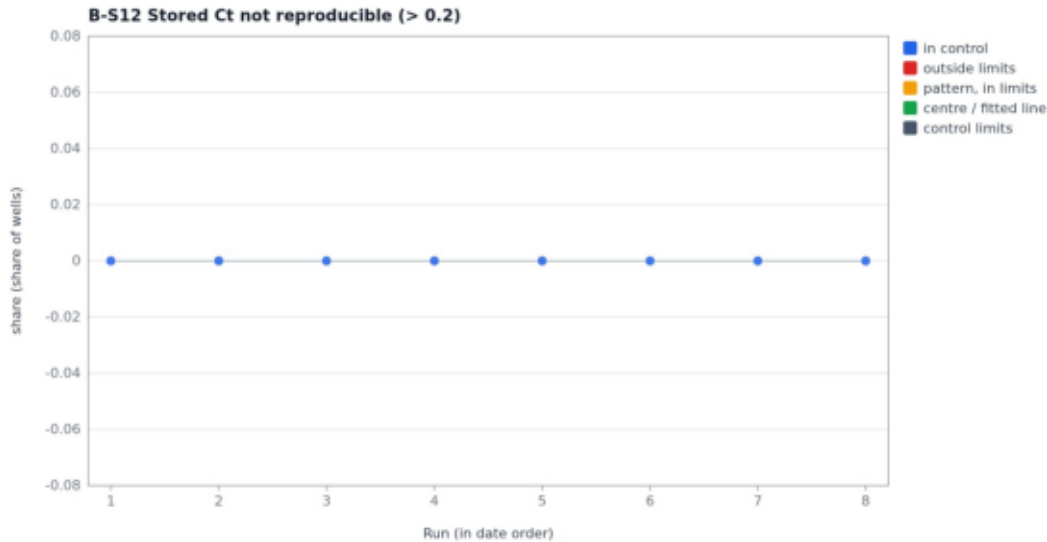
·

figure

SOUND

S21_run01.eds, S21_run02.eds

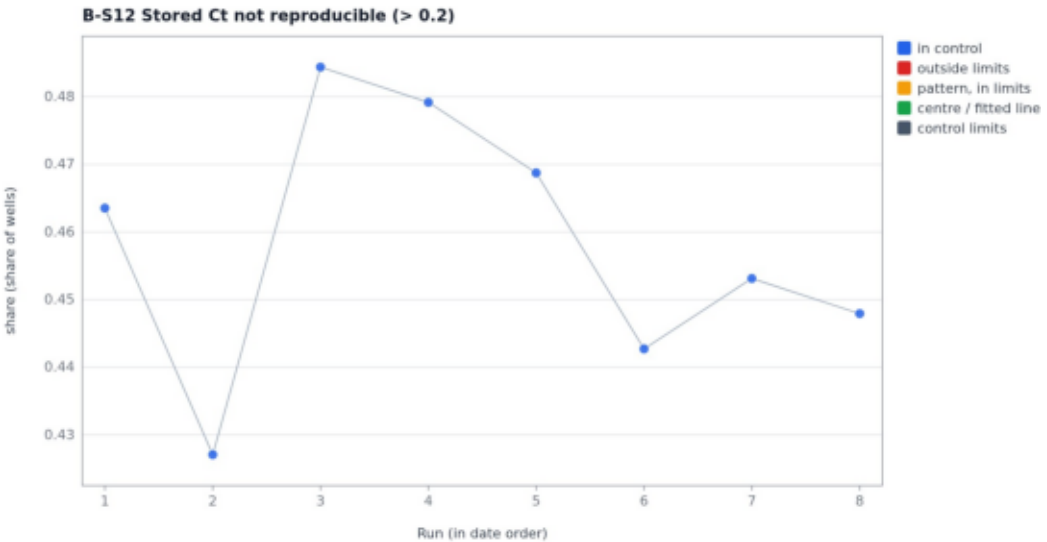
(+6 more)



DEVIATION

S21_run19.eds, S21_run20.eds

(8 files)



WHAT IT SHOWS

Non-zero requires checking algorithm, precision, stored settings and provenance; it does not establish post-analysis editing.

HOW TO READ THE TWO SIDES

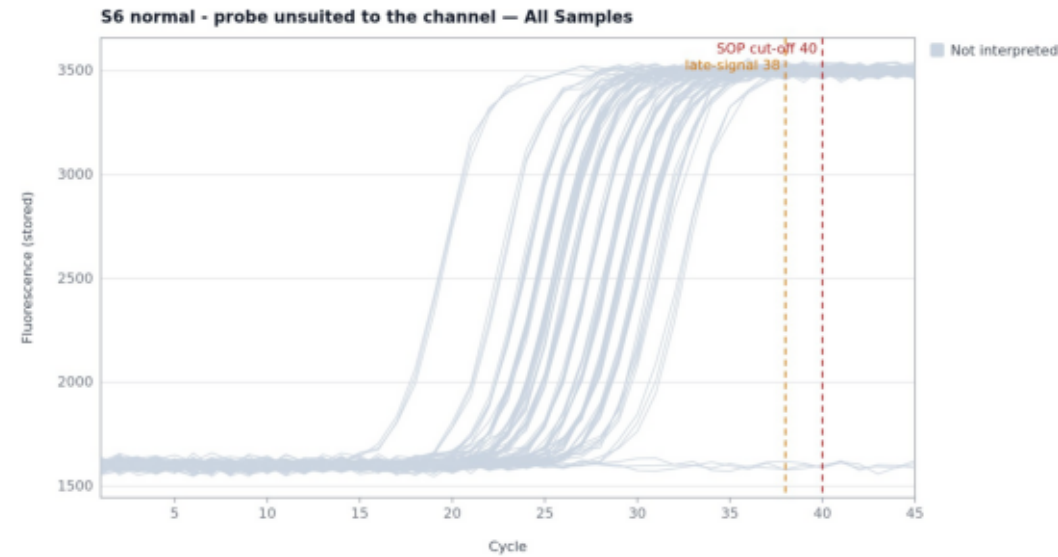
63 % of everything drawn moved between the two sides. The runs come from S21, threshold switched from automatic to manual — the first eight runs of the series against its last eight. A chart of this kind moves run to run whether or not a deviation is present, so the movement here is not itself the deviation. What matters is the level: the corpus of laboratory and vendor files puts it between 0 and 0 share of wells, around 0, and the generated history is written to sit inside that.

also moved by: S21 (52 %), S30 (3 %), S30 (3 %), S1 (0 %), S2 (0 %)

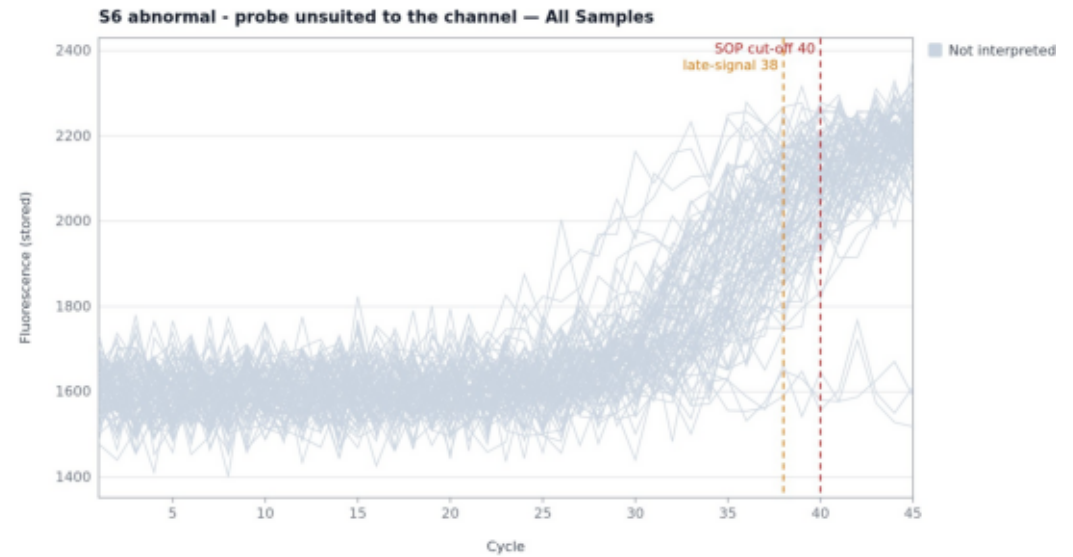
RUO NOT VALIDATED

img:curves · Fluorescence · figure

SOUND S6_normal.ixx



DEVIATION S6_abnormal.ixx

**WHAT IT SHOWS**

Stored curves of the selected target. Colour = outcome under the active SOP; dashed vertical lines = SOP Cq cut-off and late-signal limit; horizontal line = stored threshold (QuantStudio ΔRn).

WHAT CHANGED AND WHAT IT MEANS

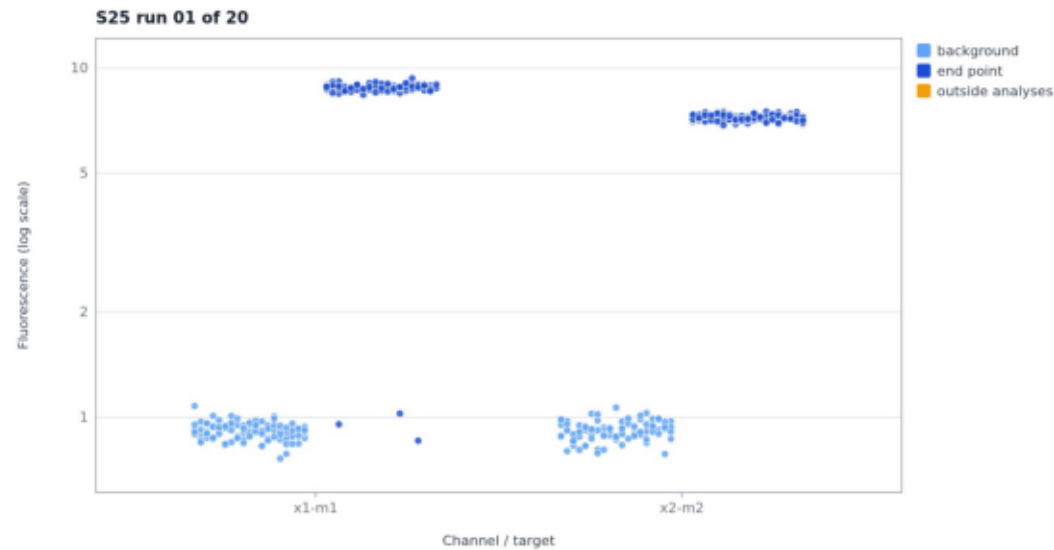
88 % of everything drawn on the figure moved between the two sides. The deviating side is the sound file beside the deviating one of S6, probe unsuited to the channel. Onset: the abnormal file from the first cycles. The signal is one-third amplitude, three times noisier, drifts, and crosses about five cycles late; plateaus are poorly formed. What to look at: Low plateau, noisy baseline and delayed threshold crossing occur together across the plate.

also moved by: S43 (81 %), S7 (79 %), S43 (75 %), S1 (72 %), S28 (70 %)

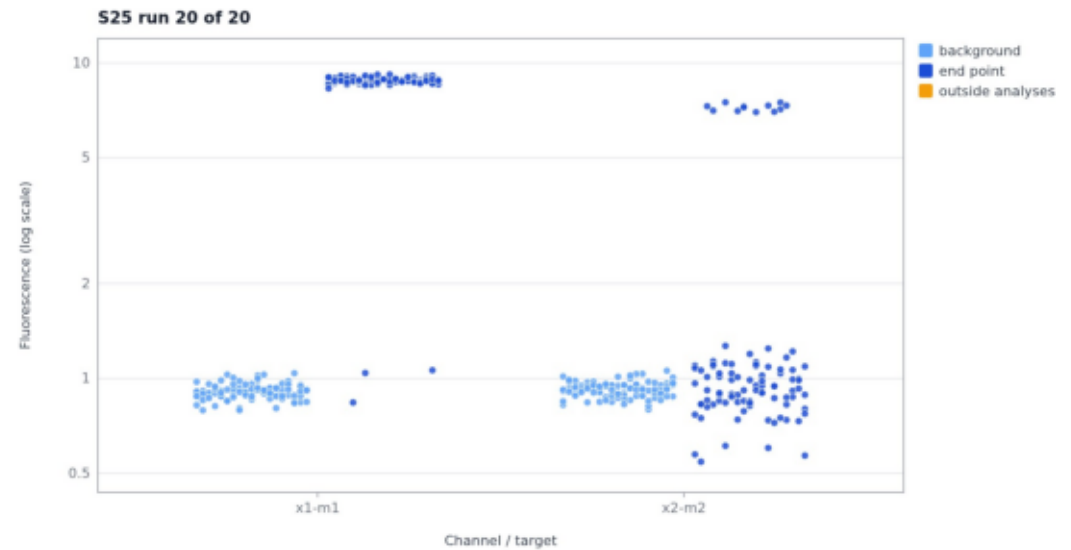
RUO NOT VALIDATED

img:levels · Fluorescence · figure

SOUND S25_run01.eds



DEVIATION S25_run20.eds

**WHAT IT SHOWS**

For each detection channel / target: background (mean of cycles 3–10) and end-point fluorescence of every well, including wells outside the analyses. A shifted background points to plate, seal or optical problems.

WHAT CHANGED AND WHAT IT MEANS

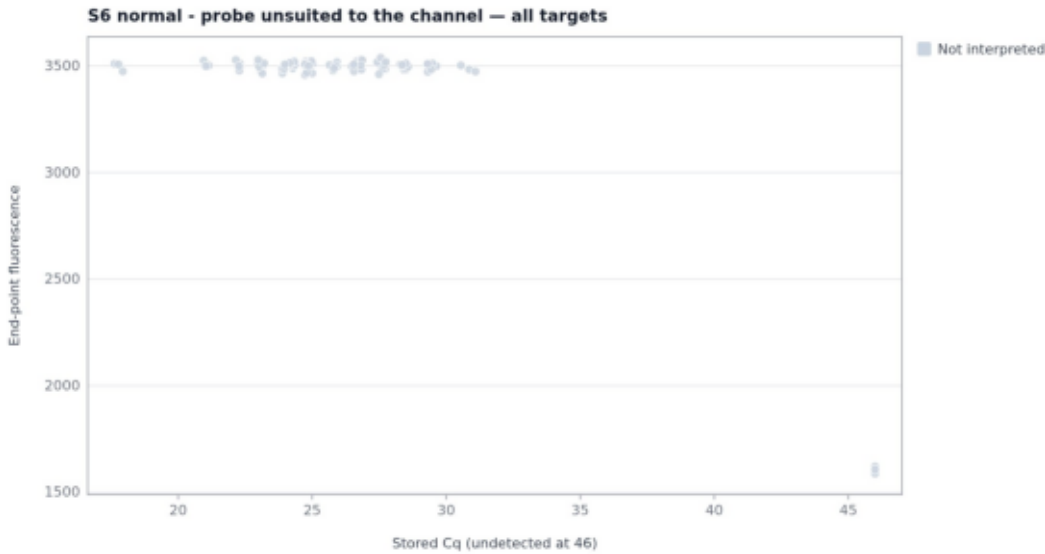
76 % of everything drawn on the figure moved between the two sides. The deviating side is the first run of the series beside its last run of S25, second target of the multiplex failing. Onset: run 09. VIC reference-gene failures begin at run 9 and reach 74 undetermined wells by run 20 while FAM remains clean. What to look at: Channel-specific failure separates the reference target from the target-A signal.

also moved by: S25 (73 %), S20 (72 %), S24 (72 %), S20 (71 %), S30 (70 %)

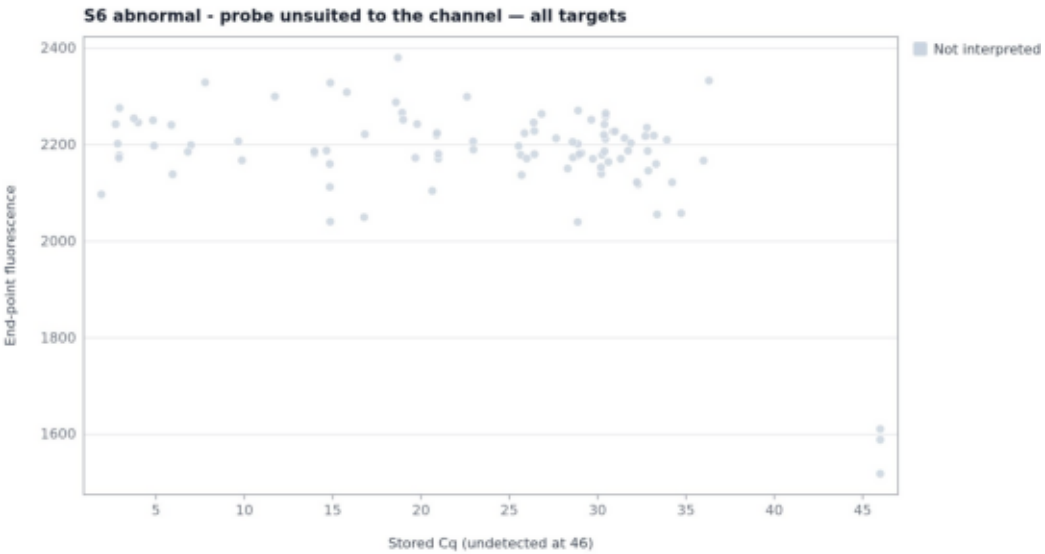
RUO NOT VALIDATED

img:endpoint · Fluorescence · figure

SOUND S6_normal.ix



DEVIATION S6_abnormal.ix



WHAT IT SHOWS

End-point fluorescence (or amplitude above background) against the stored Cq. Late, low-plateau curves separate from true amplification; undetected wells are drawn at the right edge.

WHAT CHANGED AND WHAT IT MEANS

74 % of everything drawn on the figure moved between the two sides. The deviating side is the sound file beside the deviating one of S6, probe unsuited to the channel. Onset: the abnormal file from the first cycles. The signal is one-third amplitude, three times noisier, drifts, and crosses about five cycles late; plateaus are poorly formed. What to look at: Low plateau, noisy baseline and delayed threshold crossing occur together across the plate.

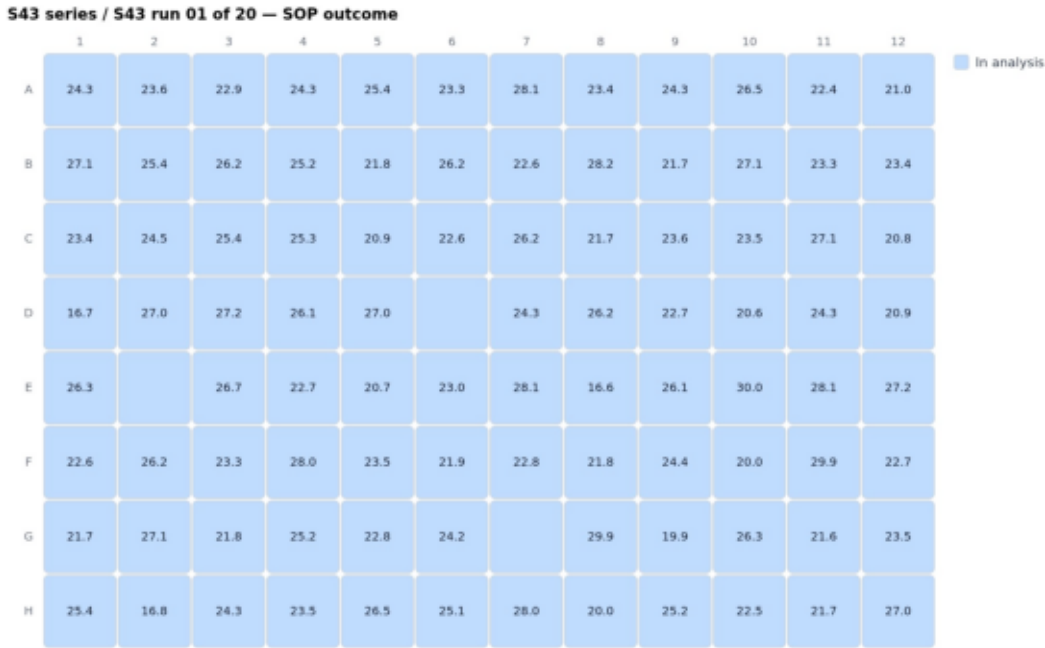
also moved by: S7 (73 %), S5 (72 %), S25 (71 %), S28 (70 %), S25 (69 %)

RUO NOT VALIDATED

img:plate · Plate · figure

SOUND S43_run01.rdm1

DEVIATION S43_run20.rdm1



S43 series / S43 run 20 of 20

WHAT IT SHOWS

Every physical position. Choose what the colour shows: SOP outcome, stored Cq, end fluorescence, background, amplitude, or analysis status (which also shows wells with signal that are in no analysis, and omitted wells).

WHAT CHANGED AND WHAT IT MEANS

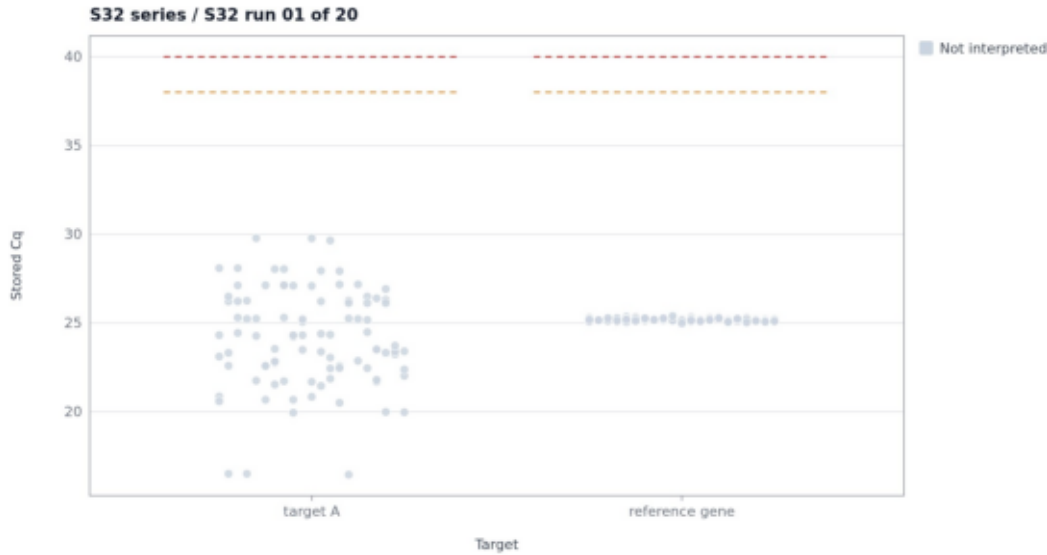
99 % of everything drawn on the figure moved between the two sides. The deviating side is the first run of the series beside its last run of S43, plate declared as free format. Onset: run 12. From run 12 `pcrFormat` becomes -1×-1 ; a plate-shaped reader keeps only 2 of 192 rows. What to look at: The format declaration changes before the apparent data loss.

also moved by: S43 (99 %), S6 (2 %), S7 (2 %), S32 (1 %), S32 (1 %)

RUO NOT VALIDATED

img:cqstrip · Results · figure

SOUND S32_run01.rdm1



DEVIATION S32_run20.rdm1



WHAT IT SHOWS

One point per well. Colour by SOP outcome or by role. Red dashed line: Cq cut-off; orange: late-signal limit (per target).

WHAT CHANGED AND WHAT IT MEANS

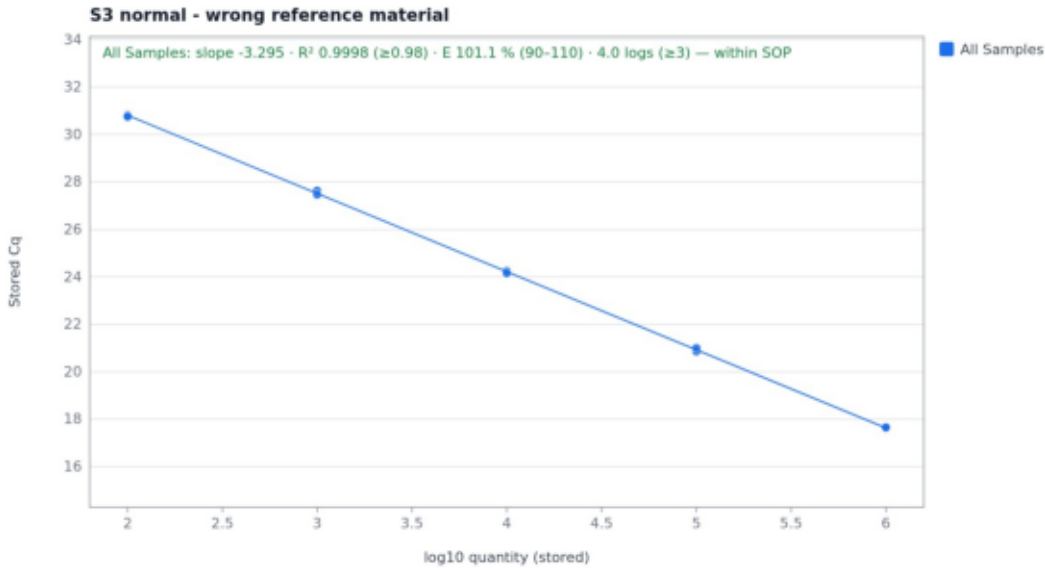
78 % of everything drawn on the figure moved between the two sides. The deviating side is the first run of the series beside its last run of S32, crossing points stop being exported. Onset: run 11. From run 11 the files carry fluorescence only and no Cq; the record changes from analysed results to raw data. What to look at: Cq rows drop to zero while `adp` fluorescence remains present.

also moved by: S32 (78 %), S25 (77 %), S25 (77 %), S26 (76 %), S22 (76 %)

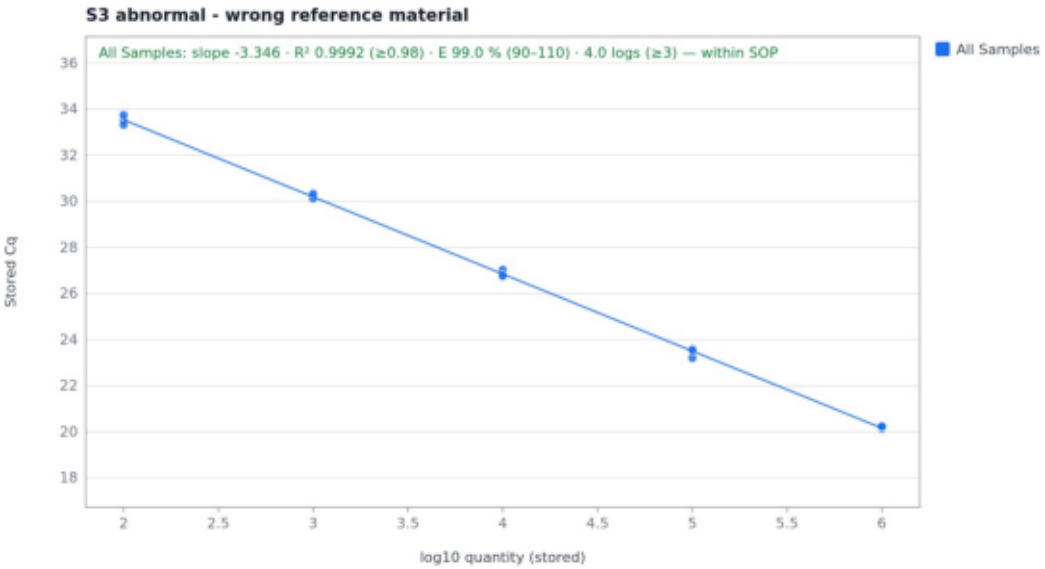
RUO NOT VALIDATED

img:stdcurve · Results · figure

SOUND S3_normal.ix



DEVIATION S3_abnormal.ix



WHAT IT SHOWS

Cq against log10 of the stored standard quantity, fitted by least squares on this page. The label shows slope, R², efficiency and range against the SOP limits; residuals are in the table.

WHAT CHANGED AND WHAT IT MEANS

75 % of everything drawn on the figure moved between the two sides. The deviating side is the sound file beside the deviating one of S3, wrong reference material. Onset: the abnormal file calibration line. The whole calibrator ladder shifts about 2.6 cycles later while spacing and efficiency remain unchanged, biasing every quantity by one factor. What to look at: The ladder is parallel to normal but displaced; inspect standard-curve intercept and calibrator Cq.

also moved by: S6 (72 %), S4 (71 %), S22 (71 %), S21 (70 %), S1 (70 %)

RUO

NOT VALIDATED

img:lc_thermal

·

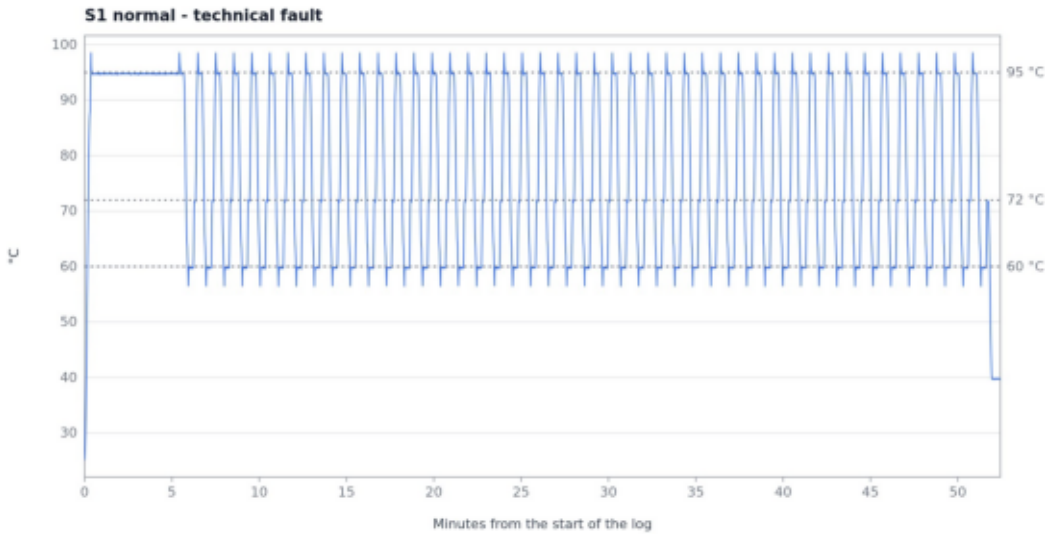
Run history

·

figure

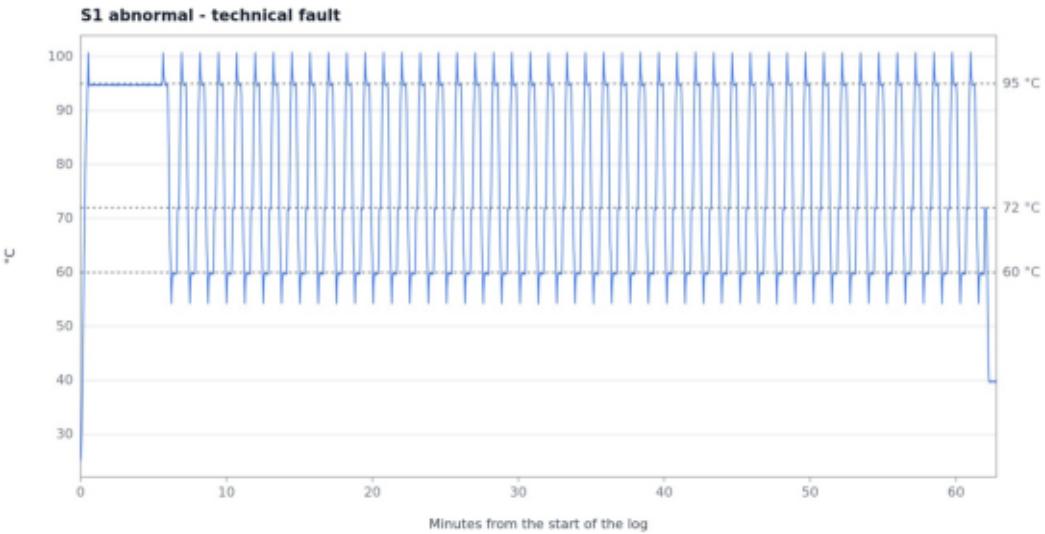
SOUND

S1_normal.ixο



DEVIATION

S1_abnormal.ixο



WHAT IT SHOWS

The block temperature the LightCycler 480 logged about every 0.41 s, decoded from the TemperatureLog inside the .ixο file, with the protocol setpoints. Heating and cooling rates, overshoot, settling and transition times on the control panel are measured on the full trace; the graph keeps the lowest and highest value of every time slice, so overshoots and short holds are not lost when it is thinned for display.

WHAT CHANGED AND WHAT IT MEANS

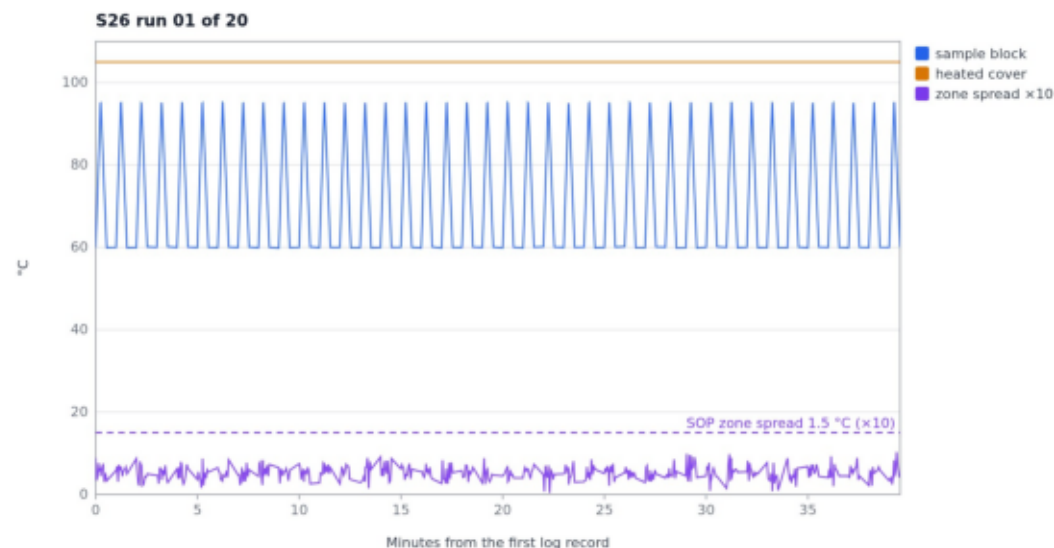
91 % of everything drawn on the figure moved between the two sides. The deviating side is the sound file beside the deviating one of S1, technical fault. Onset: cycle 29 in the abnormal file. The detector gain steps 1.55× at cycle 29 across the block; a four-cycle square artefact appears in one well and another well is dead. What to look at: All traces jump together at cycle 29; inspect the flagged trace and the missing well.

also moved by: S16 (82 %), S13 (79 %), S7 (74 %), S4 (73 %), S9 (72 %)

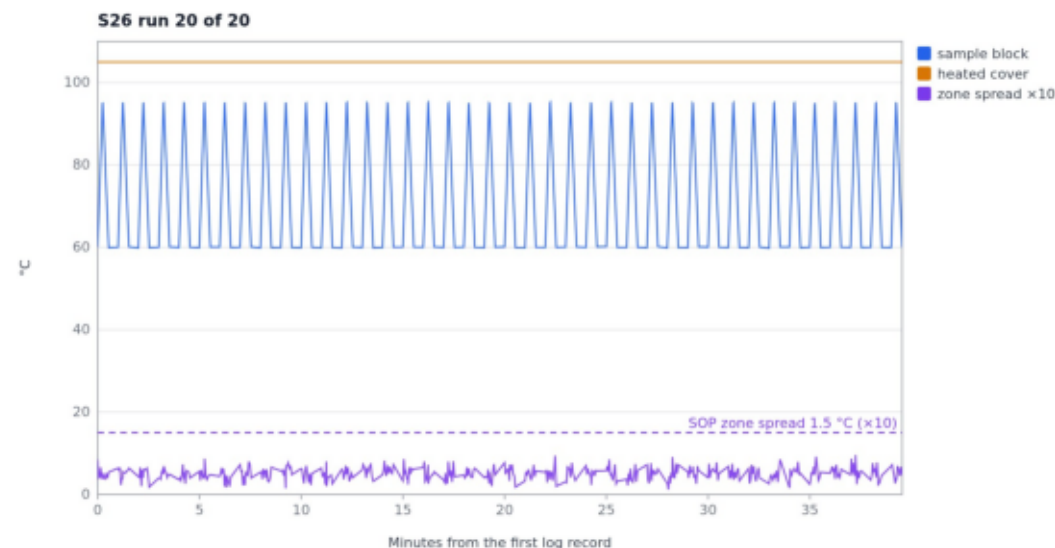
RUO NOT VALIDATED

img:temperature · Run history · figure

SOUND S26_run01.eds



DEVIATION S26_run20.eds



WHAT IT SHOWS

From the instrument log inside the .eds file: sample block mean, heated cover and the spread between block zones, against minutes from the first record. The dashed line is the SOP maximum zone spread.

WHAT CHANGED AND WHAT IT MEANS

27 % of everything drawn on the figure moved between the two sides. The deviating side is the first run of the series beside its last run of S26, relative-quantification reference changed. Onset: run 11. The $\Delta\Delta C_t$ reference changes at run 11; ΔC_q is unchanged but RQ jumps from about 0.15 to 7.05. What to look at: The discontinuity is calculation provenance, not a fluorescence change.

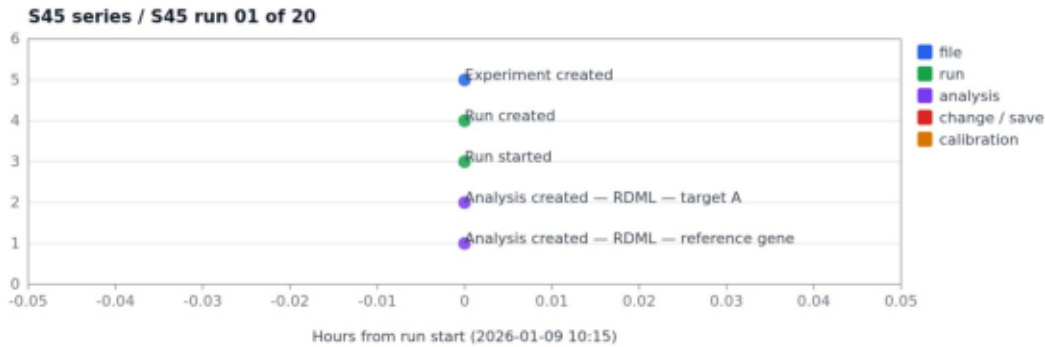
also moved by: S26 (27 %), S30 (26 %), S20 (25 %), S25 (25 %), S30 (24 %)

RUO NOT VALIDATED

img:timeline · Run history · figure

SOUND S45_run01.rdml, S45_run02.rdml (+6 more)

DEVIATION S45_run19.rdml, S45_run20.rdml (8 files)



The file carries no dated events.

WHAT IT SHOWS

Every dated event the file carries: experiment creation, run start and end, analysis creation and edits, the last recorded change (for .eds, the newest ZIP entry date is the last save), instrument log and calibrations. Events more than 7 days from the run are listed in the table and marked at the edge.

WHAT CHANGED AND WHAT IT MEANS

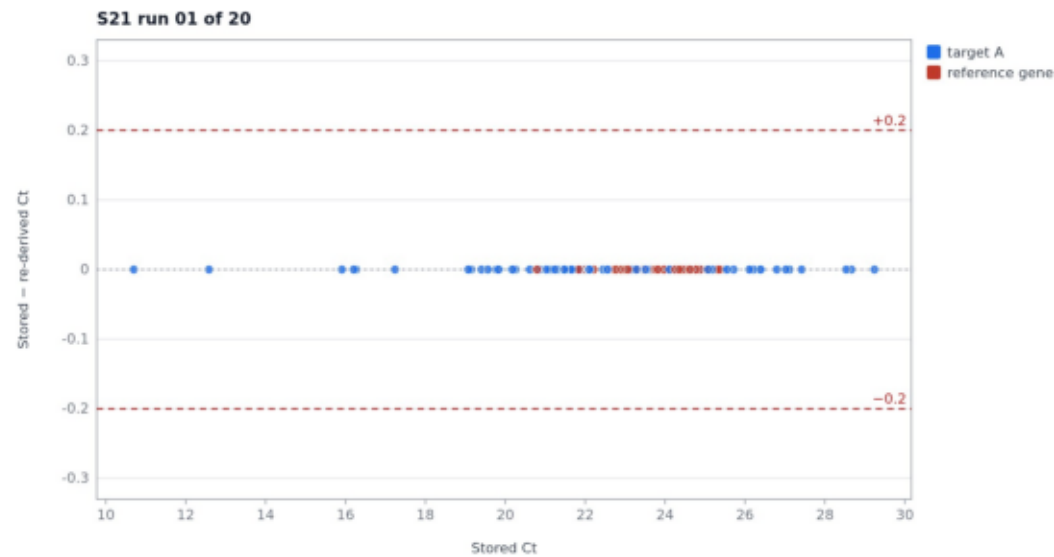
100 % of everything drawn on the figure moved between the two sides. The deviating side is the first eight runs of the series beside its last eight of S45, run date missing from selected files. Onset: runs 13, 15, 17 and 19. Four intermittent files omit `runDate`; the series cannot be ordered from file metadata alone at those points. What to look at: Missing dates recur at odd-numbered later runs rather than forming one continuous block.

also moved by: S28 (47 %), S28 (47 %), S27 (36 %), S21 (36 %), S19 (36 %)

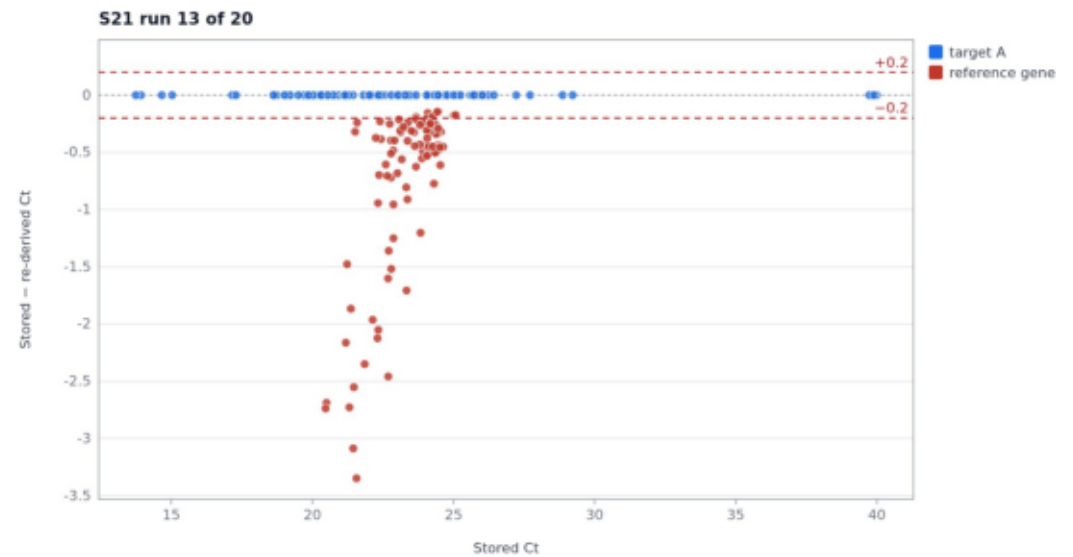
RUO NOT VALIDATED

img:recalc · Run history · figure

SOUND S21_run01.eds, S21_run02.eds (+6 more)



DEVIATION S21_run19.eds, S21_run20.eds (8 files)

**WHAT IT SHOWS**

QuantStudio only: difference between the stored Ct and the crossing of the stored ΔR_n with the stored threshold. Differences beyond ± 0.2 require an algorithm/settings/provenance review; they do not prove that stored data were edited.

WHAT CHANGED AND WHAT IT MEANS

80 % of everything drawn on the figure moved between the two sides. The deviating side is the first eight runs of the series beside its last eight of S21, threshold switched from automatic to manual. Onset: run 13. AutoCt changes from true at threshold 0.200 to false at 0.085; Cq drops about 1.2 cycles with unchanged curves. What to look at: The Cq discontinuity follows the analysis setting, not fluorescence.

also moved by: S21 (80 %), S30 (34 %), S30 (34 %), S25 (28 %), S28 (26 %)

RUO NOT VALIDATED

data:amplification_curves · Tables · table

SOUND S43_run01.rdm1, S43_run02.rdm1 (+6 more)

DEVIATION S43_run19.rdm1, S43_run20.rdm1 (8 files)

experiment	well	sample	target	cycle	fluorescence	experiment	well	sample	target	cycle	fluorescence
S43 series / S43 run01		sample 04	target A	1	0.4201	S43 series / S43 run113		calibrator level 1	target A	1	0.4182
S43 series / S43 run01		sample 04	target A	2	0.4223	S43 series / S43 run113		calibrator level 1	target A	2	0.4216
S43 series / S43 run01		sample 04	target A	3	0.4212	S43 series / S43 run113		calibrator level 1	target A	3	0.4131
S43 series / S43 run01		sample 04	target A	4	0.4229	S43 series / S43 run113		calibrator level 1	target A	4	0.4182
S43 series / S43 run01		sample 04	target A	5	0.431	S43 series / S43 run113		calibrator level 1	target A	5	0.4348
S43 series / S43 run01		sample 04	target A	6	0.4166	S43 series / S43 run113		calibrator level 1	target A	6	0.4176
S43 series / S43 run01		sample 04	target A	7	0.4104	S43 series / S43 run113		calibrator level 1	target A	7	0.4049
S43 series / S43 run01		sample 04	target A	8	0.4215	S43 series / S43 run113		calibrator level 1	target A	8	0.4104
S43 series / S43 run01		sample 04	target A	9	0.4304	S43 series / S43 run113		calibrator level 1	target A	9	0.422
S43 series / S43 run01		sample 04	target A	10	0.4125	S43 series / S43 run113		calibrator level 1	target A	10	0.4223

WHAT IT SHOWS

The fluorecence of every well, one row per cycle.

WHAT CHANGED AND WHAT IT MEANS

60800 more or fewer rows and 62.4 % of the shared cells differ between the two sides. The deviating side is the first eight runs of the series beside its last eight of S43, plate declared as free format. Onset: run 12. From run 12 `pcrFormat` becomes -1×-1 ; a plate-shaped reader keeps only 2 of 192 rows. What to look at: The format declaration changes before the apparent data loss.

also moved by: S13 (50 %), S43 (50 %), S28 (50 %), S29 (50 %), S36 (50 %)

RUO NOT VALIDATED

data:control_charts · Tables · table

SOUND S43_run01.rdml, S43_run02.rdml (+6 more)

DEVIATION S43_run19.rdml, S43_run20.rdml (8 files)

series	category	target	experiment	date	tick	cq	series	category	target	experiment	date	tick	cq
calibrator levelStandard		reference gene	S43 series / S43026-01-09T10:15:00			25.13066666666666	calibrator levelStandard		reference gene	S43 series / S43026-06-02T10:15:00			25.077
calibrator levelStandard		reference gene	S43 series / S43026-02-21T11:15:00			25.16466666666665	calibrator levelStandard		target A	S43 series / S43026-06-02T10:15:00			16.467
calibrator levelStandard		reference gene	S43 series / S43026-03-02T12:15:00			25.172	calibrator levelStandard		reference gene	S43 series / S43026-09-25T09:15:00			25.348
calibrator levelStandard		reference gene	S43 series / S43026-02-14T09:15:00			25.16933333333333	calibrator levelStandard		target A	S43 series / S43026-09-25T09:15:00			29.509
calibrator levelStandard		reference gene	S43 series / S43026-05-26T10:15:00			25.099	no-template control		reference gene	S43 series / S43026-09-13T12:15:00			25.23
calibrator levelStandard		reference gene	S43 series / S43026-06-10T11:15:00			25.18966666666668	no-template control		target A	S43 series / S43026-09-13T12:15:00			
calibrator levelStandard		reference gene	S43 series / S43026-07-22T12:15:00			25.16033333333333							
calibrator levelStandard		reference gene	S43 series / S43026-08-03T09:15:00			25.248							
calibrator levelStandard		target A	S43 series / S43026-01-09T10:15:00			16.715999999999998							
calibrator levelStandard		target A	S43 series / S43026-02-21T11:15:00			16.563999999999997							

WHAT IT SHOWS

The plotted points of the control series with their limits.

WHAT CHANGED AND WHAT IT MEANS

106 more or fewer rows and 75.0 % of the shared cells differ between the two sides. The deviating side is the first eight runs of the series beside its last eight of S43, plate declared as free format. Onset: run 12. From run 12 `pcrFormat` becomes -1×-1 ; a plate-shaped reader keeps only 2 of 192 rows. What to look at: The format declaration changes before the apparent data loss.

also moved by: S43 (70 %), S32 (63 %), S32 (63 %), S36 (53 %), S36 (52 %)

RUO NOT VALIDATED

data:experiment_meta · Tables · table

SOUND S14_run01.ix

file	platform	integrity	source_uid	source_crc32	source_sha256	source_bytes
S14_run01.ix	LightCycler 480 container checks	57092833DCBB4B204D5702E9B			1413760d1c75295f0756854da	

DEVIATION S14_run20.ix

file	platform	integrity	source_uid	source_crc32	source_sha256	source_bytes
S14_run20.ix	LightCycler 480 container checks	0706387DE5D2418687654421			8bd46f6365b502ad31095493	

WHAT IT SHOWS

Metadata fields read from every loaded experiment.

WHAT CHANGED AND WHAT IT MEANS

0 more or fewer rows and 45.2 % of the shared cells differ between the two sides. The deviating side is the first run of the series beside its last run of S14, position effect on the plate. Onset: every run; visible after aggregation. Across freshly shuffled maps, edge wells sit about +0.26, the last column +0.42 and interior wells −0.16 relative to each sample’s own replicates. What to look at: The effect is spatial and consistent even though sample identities move between positions.

also moved by: S9 (42 %), S10 (42 %), S11 (42 %), S12 (42 %), S13 (42 %)

RUO NOT VALIDATED

data:experiment_settings · Tables · table

SOUND S10_run01.ixo, S10_run02.ixo (+6 more)

setting	value	source
Analysis modules	Absolute Quantification	8 analysis object(s) in
Detection channels	465-510 465-510	HTCDetectionFormat
Plate	8 x 12 (96)	HTCBlockType RowCount /
Study result	Crossing Point	quantification results p
Amplification curves	768 linked; 768 acquired	AcquisitionStore
Cycles acquired	45	AcquisitionStore
Fluorescence scaling fac	10000	AcquisitionStore THTCFlo
Stored results	768 quantification	per-analysis result obje
Sample roles	Unknown, Positive contro	QuantSampleTypeProperty,
Replicates	identical sample names	no replicate grouping re

DEVIATION S10_run19.ixo, S10_run20.ixo (8 files)

setting	value	source
Analysis modules	Absolute Quantification	8 analysis object(s) in
Detection channels	465-510 465-510	HTCDetectionFormat
Plate	8 x 12 (96)	HTCBlockType RowCount /
Study result	Crossing Point	quantification results p
Amplification curves	768 linked; 768 acquired	AcquisitionStore
Cycles acquired	45	AcquisitionStore
Fluorescence scaling fac	10000	AcquisitionStore THTCFlo
Stored results	768 quantification	per-analysis result obje
Sample roles	Positive control, Standa	QuantSampleTypeProperty,
Replicates	identical sample names	no replicate grouping re

WHAT IT SHOWS

Every setting read out of the container, with the property it came from.

WHAT CHANGED AND WHAT IT MEANS

1 more or fewer rows and 90.7 % of the shared cells differ between the two sides. The deviating side is the first eight runs of the series beside its last eight of S10, efficiency falling over the series. Onset: run 06. From run 6 the calibration spacing and amplitude deteriorate; stored efficiency falls from about 1.01 toward 0.76 by run 20. What to look at: The first persistent downward change is run 6; later runs widen the separation from baseline.

also moved by: S45 (79 %), S34 (73 %), S10 (59 %), S11 (59 %), S12 (59 %)

RUO NOT VALIDATED

data:forensic_log · Tables · table

SOUND S28_run01.eds, S28_run02.eds (+6 more) DEVIATION S28_run19.eds, S28_run20.eds (8 files)

experiment	severity	area	finding	evidence	experiment	severity	area	finding	evidence
S28 run 01 of 20	info	Recovery	QuantStudio experiment8#12; 2 analyses; 192 qu		S28 run 13 of 20	info	Recovery	QuantStudio experiment8#12; 2 analyses; 192 qu	
S28 run 01 of 20	review	Sample role	15 instrument-versus-nA05 calibrator level 2:		S28 run 13 of 20	review	Sample role	15 instrument-versus-nB05 calibrator level 3:	
S28 run 01 of 20	review	Assay controls	3 NTC/blank result(s) A03 no-template control		S28 run 13 of 20	info	Integrity	ZIP CRC-32 valid for all	
S28 run 01 of 20	info	Integrity	ZIP CRC-32 valid for all		S28 run 13 of 20	info	Integrity	Tamper MD5 matches the1E81F1C21C294EC2CE16234B	
S28 run 01 of 20	info	Integrity	Tamper MD5 matches the4861C275010F58715D7D13DF		S28 run 13 of 20	info	Ct traceability	Stored Ct re-derived f109 wells compared; the	
S28 run 01 of 20	review	Assay controls	1 NTC result(s) with aB11 target A 36.00		S28 run 13 of 20	error	Calibration	1 calibration(s) expir00I 2026-01-05 00:00:00	
S28 run 01 of 20	review	Replicates	sample 25 / reference gange 23.57–25.11; wells		S28 run 13 of 20	info	Plate setup	Rn is normalised to the	
S28 run 01 of 20	review	Replicates	no-template control / tange 14.82–36.00; wells		S28 run 13 of 20	info	Standard curve	target A: slope -3.3079,intercept 35.297; 15 s	
S28 run 01 of 20	review	Replicates	no-template control / range 23.56–24.67; wells		S28 run 13 of 20	info	Flags	32 stored flag(s) on 3BADROX, ROXL0W	
S28 run 01 of 20	review	Replicates	sample 11 / target A: fange 20.88–21.92; wells		S28 run 13 of 20	info	Instrument log	480 temperature recordsyn 2026-06-02 10:15:12	

WHAT IT SHOWS

Every finding with its area, severity and the evidence behind it.

WHAT CHANGED AND WHAT IT MEANS

304 more or fewer rows and 94.0 % of the shared cells differ between the two sides. The deviating side is the first eight runs of the series beside its last eight of S28, calibration allowed to expire. Onset: the expiry date inside the series; flags from run 13. The pure-dye calibration expiry falls inside the run series; passive-reference flags appear from run 13 and increase thereafter. What to look at: The provenance date precedes the first sustained flags.

also moved by: S20 (86 %), S28 (82 %), S19 (81 %), S43 (80 %), S23 (74 %)

RUO

NOT VALIDATED

rec:instrument_records · Instrument records · table

SOUND

S28_run01.eds, S28_run02.eds

(+6 more)

run	experiment	table	field	value	check	result
1	S28_run 01 of 20	metadata	name	S28_run 01 of 20		
1	S28_run 01 of 20	metadata	UID			
1	S28_run 01 of 20	metadata	Created	2026-01-09T09:40:00+00:0		
1	S28_run 01 of 20	metadata	CreatedByName	analyst A		
1	S28_run 01 of 20	metadata	LastModified	2026-01-15T22:15:00+00:0		
1	S28_run 01 of 20	metadata	LastModifiedByName			
1	S28_run 01 of 20	metadata	SWVersion			
1	S28_run 01 of 20	metadata	AppliedTemplates			
1	S28_run 01 of 20	metadata	State	COMPLETE		
1	S28_run 01 of 20	metadata	Notes	generated demonstration		

DEVIATION

S28_run19.eds, S28_run20.eds

(8 files)

run	experiment	table	field	value	check	result
1	S28_run 13 of 20	metadata	name	S28_run 13 of 20		
1	S28_run 13 of 20	metadata	UID			
1	S28_run 13 of 20	metadata	Created	2026-06-02T09:40:00+00:0		
1	S28_run 13 of 20	metadata	CreatedByName	analyst A		
1	S28_run 13 of 20	metadata	LastModified	2026-06-08T22:15:00+00:0		
1	S28_run 13 of 20	metadata	LastModifiedByName			
1	S28_run 13 of 20	metadata	SWVersion			
1	S28_run 13 of 20	metadata	AppliedTemplates			
1	S28_run 13 of 20	metadata	State	COMPLETE		
1	S28_run 13 of 20	metadata	Notes	generated demonstration		

WHAT IT SHOWS

Metadata, protocol, channels, subsets, analyses and statistics.

WHAT CHANGED AND WHAT IT MEANS

8 more or fewer rows and 5.1 % of the shared cells differ between the two sides. The deviating side is the first eight runs of the series beside its last eight of S28, calibration allowed to expire. Onset: the expiry date inside the series; flags from run 13. The pure-dye calibration expiry falls inside the run series; passive-reference flags appear from run 13 and increase thereafter. What to look at: The provenance date precedes the first sustained flags.

also moved by: S28 (5 %), S20 (4 %), S20 (4 %), S30 (4 %), S21 (4 %)

RUO NOT VALIDATED

data:multicomponent_raw · Tables · table

SOUND S25_run01.eds, S25_run02.eds (+6 more) DEVIATION S25_run19.eds, S25_run20.eds (8 files)

experiment	well	sample	cycle	multicomponent_FAM	multicomponent_WT	multicomponent_ROX	experiment	well	sample	cycle	multicomponent_FAM	multicomponent_WT	multicomponent_ROX
S25 run 01 of 20	A01	calibrator level1	3	158377.5128	166709.6757	192310.6092	S25 run 13 of 20	A01	S01	1	162117.9276	167427.8805	168327.1384
S25 run 01 of 20	A01	calibrator level2	3	126938.8109	147813.5222	215820.3356	S25 run 13 of 20	A01	S01	2	122693.1347	156453.6928	198609.5892
S25 run 01 of 20	A01	calibrator level3	3	184449.1107	175020.3413	224184.4009	S25 run 13 of 20	A01	S01	3	126328.6746	235696.074	149000.5832
S25 run 01 of 20	A01	calibrator level4	3	132601.7778	170032.2352	156070.2146	S25 run 13 of 20	A01	S01	4	170774.7663	128793.1637	173760.0686
S25 run 01 of 20	A01	calibrator level5	3	137526.6263	129766.1002	164476.9127	S25 run 13 of 20	A01	S01	5	171253.3531	169657.5201	175236.8029
S25 run 01 of 20	A01	calibrator level6	3	163389.8937	176299.6823	224370.3336	S25 run 13 of 20	A01	S01	6	196483.1743	171210.2077	166739.0819
S25 run 01 of 20	A01	calibrator level7	3	139168.0628	173633.2741	131872.5194	S25 run 13 of 20	A01	S01	7	117321.1806	232391.2964	187193.9998
S25 run 01 of 20	A01	calibrator level8	3	192108.1894	181505.7407	188293.542	S25 run 13 of 20	A01	S01	8	153052.9159	126910.9623	197746.8032
S25 run 01 of 20	A01	calibrator level9	3	133932.5211	177508.0909	167798.0884	S25 run 13 of 20	A01	S01	9	183804.717	171638.9401	191141.017
S25 run 01 of 20	A01	calibrator level10	3	188631.5528	151760.9342	174660.296	S25 run 13 of 20	A01	S01	10	190076.9802	178287.3901	175188.2189

WHAT IT SHOWS

Raw multicomponent signal rows from exchange containers.

WHAT CHANGED AND WHAT IT MEANS

0 more or fewer rows and 74.7 % of the shared cells differ between the two sides. The deviating side is the first eight runs of the series beside its last eight of S25, second target of the multiplex failing. Onset: run 09. VIC reference-gene failures begin at run 9 and reach 74 undetermined wells by run 20 while FAM remains clean. What to look at: Channel-specific failure separates the reference target from the target-A signal.

also moved by: S22 (75 %), S30 (75 %), S26 (75 %), S19 (75 %), S23 (75 %)

RUO NOT VALIDATED

data:plate_map · Tables · table

SOUND S43_run01.rdm1, S43_run02.rdm1 (+6 more)

DEVIATION S43_run19.rdm1, S43_run20.rdm1 (8 files)

experiment	source_uid	source_crc32	well	well_lims	position	sample	experiment	source_uid	source_crc32	well	well_lims	position	sample
S43 series / S43 run 01			A01	A1	1	sample 04	S43 series / S43 run 13			A01	A1	1	calibrator level 1
S43 series / S43 run 01			A02	A2	2	sample 03	S43 series / S43 run 14			A01	A1	1	sample 03
S43 series / S43 run 01			A03	A3	3	run control	S43 series / S43 run 15			A01	A1	1	sample 10
S43 series / S43 run 01			A04	A4	4	sample 04	S43 series / S43 run 16			A01	A1	1	sample 12
S43 series / S43 run 01			A05	A5	5	sample 14	S43 series / S43 run 17			A01	A1	1	sample 06
S43 series / S43 run 01			A06	A6	6	calibrator level 3	S43 series / S43 run 18			A01	A1	1	sample 04
S43 series / S43 run 01			A07	A7	7	sample 17	S43 series / S43 run 19			A01	A1	1	no-template control
S43 series / S43 run 01			A08	A8	8	sample 12	S43 series / S43 run 20			A01	A1	1	calibrator level 5
S43 series / S43 run 01			A09	A9	9	sample 04							
S43 series / S43 run 01			A10	A10	10	calibrator level 4							

WHAT IT SHOWS

The plate as the file describes it: position, sample, target and role.

WHAT CHANGED AND WHAT IT MEANS

760 more or fewer rows and 38.7 % of the shared cells differ between the two sides. The deviating side is the first eight runs of the series beside its last eight of S43, plate declared as free format. Onset: run 12. From run 12 `pcrFormat` becomes -1×-1 ; a plate-shaped reader keeps only 2 of 192 rows. What to look at: The format declaration changes before the apparent data loss.

also moved by: S11 (38 %), S13 (38 %), S10 (37 %), S15 (37 %), S16 (37 %)

RUO NOT VALIDATED

data:runs_and_qc · Tables · table

SOUND S28_run01.eds

experiment	date	source_uid	source_crc32	operator	instrument	plate
S28 run 01 of 20	2026-01-09T10:15:00+00:0		9984998a	analyst A	QuantStudio	8x12

DEVIATION S28_run20.eds

experiment	date	source_uid	source_crc32	operator	instrument	plate
S28 run 20 of 20	2026-08-25T09:15:00+00:0		bf25c571	analyst D	QuantStudio	8x12

WHAT IT SHOWS

One row per run with its profile answer and the reason behind it.

WHAT CHANGED AND WHAT IT MEANS

0 more or fewer rows and 52.9 % of the shared cells differ between the two sides. The deviating side is the first run of the series beside its last run of S28, calibration allowed to expire. Onset: the expiry date inside the series; flags from run 13. The pure-dye calibration expiry falls inside the run series; passive-reference flags appear from run 13 and increase thereafter. What to look at: The provenance date precedes the first sustained flags.

also moved by: S28 (47 %), S43 (41 %), S43 (41 %), S5 (35 %), S14 (35 %)

RUO NOT VALIDATED

data:sop_criteria · Tables · table

SOUND S1_normal.ix

DEVIATION S1_abnormal.ix

experiment	run_status	criterion	sop_action	status	evidence	sop_profile	experiment	run_status	criterion	sop_action	status	evidence	sop_profile
S1 normal - technical fault not applicable	Method applicability solved	n/a	run name S1 normal screening – SOP-06	S1 abnormal - technical fault not applicable	Method applicability solved	n/a	run name S1 abnormal screening – SOP-06						

WHAT IT SHOWS

One row per evaluated profile criterion.

WHAT CHANGED AND WHAT IT MEANS

0 more or fewer rows and 33.3 % of the shared cells differ between the two sides. The deviating side is the sound file beside the deviating one of S1, technical fault. Onset: cycle 29 in the abnormal file. The detector gain steps 1.55× at cycle 29 across the block; a four-cycle square artefact appears in one well and another well is dead. What to look at: All traces jump together at cycle 29; inspect the flagged trace and the missing well.

also moved by: S2 (33 %), S3 (33 %), S4 (33 %), S5 (33 %), S6 (33 %)

RUO NOT VALIDATED

data:sop_runs · Tables · table

SOUND S1_normal.ix

experiment	platform	status	rejected	review	positive	negative
S1 normal - technical	Hydra	Cler 480	Profile not applicable	—	0	0

DEVIATION S1_abnormal.ix

experiment	platform	status	rejected	review	positive	negative
S1 abnormal - technical	Hydra	Cler 480	Profile not applicable	—	0	0

WHAT IT SHOWS

One row per run with profile status and counts.

WHAT CHANGED AND WHAT IT MEANS

0 more or fewer rows and 26.3 % of the shared cells differ between the two sides. The deviating side is the sound file beside the deviating one of S1, technical fault. Onset: cycle 29 in the abnormal file. The detector gain steps 1.55× at cycle 29 across the block; a four-cycle square artefact appears in one well and another well is dead. What to look at: All traces jump together at cycle 29; inspect the flagged trace and the missing well.

also moved by: S2 (26 %), S3 (26 %), S4 (26 %), S5 (26 %), S6 (26 %)

RUO NOT VALIDATED

data:statistics · Tables · table

SOUND S1_normal.ix

DEVIATION S1_abnormal.ix

experiment	analysis	target	results	amplified	uncertain	missing_curves	experiment	analysis	target	results	amplified	uncertain	missing_curves
S1 normal - technical	Misc data	2nd Detiva	96	93	0	0	S1 abnormal - technical	Misc data	2nd Detiva	96	92	0	0

WHAT IT SHOWS

Per target and group: counts, mean Cq, spread and detection.

WHAT CHANGED AND WHAT IT MEANS

0 more or fewer rows and 73.7 % of the shared cells differ between the two sides. The deviating side is the sound file beside the deviating one of S1, technical fault. Onset: cycle 29 in the abnormal file. The detector gain steps 1.55× at cycle 29 across the block; a four-cycle square artefact appears in one well and another well is dead. What to look at: All traces jump together at cycle 29; inspect the flagged trace and the missing well.

also moved by: S5 (74 %), S2 (63 %), S3 (63 %), S4 (63 %), S6 (63 %)

RUO NOT VALIDATED

data:cq_values · Tables · table

SOUND S19_run01.eds

DEVIATION S19_run20.eds

experiment	source_uid	source_crc32	well	well_lims	position	sample	experiment	source_uid	source_crc32	well	well_lims	position	sample
S19 run 01 of 20	194b2b51	A01	A1	1	sample 17		S19 run 20 of 20	624f0b06	A01	A1	1	sample 08	
S19 run 01 of 20	194b2b51	A01	A1	1	sample 17		S19 run 20 of 20	624f0b06	A01	A1	1	sample 08	
S19 run 01 of 20	194b2b51	A02	A2	2	no-template control	S19 run 20 of 20	624f0b06	A02	A2	2	sample 14		
S19 run 01 of 20	194b2b51	A02	A2	2	no-template control	S19 run 20 of 20	624f0b06	A02	A2	2	sample 14		
S19 run 01 of 20	194b2b51	A03	A3	3	sample 05	S19 run 20 of 20	624f0b06	A03	A3	3	sample 25		
S19 run 01 of 20	194b2b51	A03	A3	3	sample 05	S19 run 20 of 20	624f0b06	A03	A3	3	sample 25		
S19 run 01 of 20	194b2b51	A04	A4	4	sample 15	S19 run 20 of 20	624f0b06	A04	A4	4	sample 15		
S19 run 01 of 20	194b2b51	A04	A4	4	sample 15	S19 run 20 of 20	624f0b06	A04	A4	4	sample 15		
S19 run 01 of 20	194b2b51	A05	A5	5	sample 12	S19 run 20 of 20	624f0b06	A05	A5	5	sample 04		
S19 run 01 of 20	194b2b51	A05	A5	5	sample 12	S19 run 20 of 20	624f0b06	A05	A5	5	sample 04		

WHAT IT SHOWS

One row per stored result with well, sample, target, role and call.

WHAT CHANGED AND WHAT IT MEANS

0 more or fewer rows and 31.5 % of the shared cells differ between the two sides. The deviating side is the first run of the series beside its last run of S19, passive reference falling away. Onset: run 05. ROX declines from run 5 onward to about 55% below baseline by run 20; ROXLOW/BADROX flags and replicate spread increase with it. What to look at: The passive-reference trend starts before the spread worsens; use ROX and flags as leading evidence.

also moved by: S25 (31 %), S19 (31 %), S25 (30 %), S23 (29 %), S23 (29 %)

the page draws these, but nothing in the present files feeds them — its own message is quoted

CATALOGUE ENTRY

img:x_control	Control Cq across runs with SOP ra
img:x_control_thermal	Control Cq versus cooling rate
img:x_control_age	Control extract age versus Cq
img:x_control_batches	Control extraction batches
img:x_accept	Run acceptance grid
img:x_duration	Run duration and instrument use
img:x_outcomes	SOP outcomes per run
cc:calib_background	Background calibration: mean offse
cc:chan_switch	Channel switch time
cc:encoder	Filter-wheel position offset
cc:cover_down	Heated cover lowering time
cc:cover_up	Heated cover raising time
cc:cover_temp	Heated cover temperature
cc:run_minutes	Run duration
cc:calib_uniformity	Uniformity calibration: edge-to-ce
cc:calib_roi	Well-position (ROI) calibration: s
cc:ic_shift	IC shift from fixed reference
cc:late_rate	Late signal / stochastic-zone resu
cc:low_conf	Low Cq confidence
cc:ntc_rate	Negative controls by process stage
cc:pc_cq	Positive control Cq (Levey-Jenning
cc:pc_amplitude	Probe signal strength (positive co
cc:ntc_op	Contamination rate by operator (fu
cc:copied	Copied-data events
cc:action_op	Repeat / inconclusive / invalid ra
cc:rep_sd	Replicate precision (pooled SD, Cq

WHAT THE PAGE SAYS

The chosen control was not found in the loaded runs.
No eligible controls: configure positive-control names, target and valid extraction da
No eligible controls: configure positive-control names, target and valid extraction da
No eligible controls: configure positive-control names, target and valid extraction da
drawn on both sides, but no scenario moves it by more than 20 % of what it draws
drawn on both sides, but no scenario moves it by more than 2 % of what it draws
drawn on both sides, but no scenario moves it by more than 6 % of what it draws
No data for this chart in the loaded runs or the imported history.
No data for this chart in the loaded runs or the imported history.
No data for this chart in the loaded runs or the imported history.
No data for this chart in the loaded runs or the imported history.
No data for this chart in the loaded runs or the imported history.
drawn on both sides, but no scenario moves it by more than 0 % of what it draws
No data for this chart in the loaded runs or the imported history.
No data for this chart in the loaded runs or the imported history.
No data for this chart in the loaded runs or the imported history.
No data for this chart in the loaded runs or the imported history.
drawn on both sides, but no scenario moves it by more than 0 % of what it draws
No data for this chart in the loaded runs or the imported history.
No data for this chart in the loaded runs or the imported history.
No data for this chart in the loaded runs or the imported history.
No operator data for this chart.
drawn on both sides, but no scenario moves it by more than 4 % of what it draws
No operator data for this chart.
No data for this chart in the loaded runs or the imported history.

the page draws these, but nothing in the present files feeds them — its own message is quoted

CATALOGUE ENTRY		WHAT THE PAGE SAYS
cc:rowcol	Row / column pattern	No data for this chart in the loaded runs or the imported history.
cc:call_disagree	Stored calls that disagree with th	No data for this chart in the loaded runs or the imported history.
cc:log_errors	Command errors in the instrument l	No data for this chart in the loaded runs or the imported history.
cc:handoff_min	Data hand-off time	No data for this chart in the loaded runs or the imported history.
cc:roi_ms	Image analysis time	No data for this chart in the loaded runs or the imported history.
cc:img_proc	Image → processing delay	No data for this chart in the loaded runs or the imported history.
cc:invalid_acq	Invalid fluorescence readings	No data for this chart in the loaded runs or the imported history.
cc:catch_up	Scheduler falling behind	No data for this chart in the loaded runs or the imported history.
cc:templog_late	Temperature-log intervals over 500	No data for this chart in the loaded runs or the imported history.
img:unanalysed	Signal in wells that are not in an	No rising curve outside the analyses in this run.
img:ic	Internal control / reference Cq pe	No target is marked as internal control or reference in the SOP (section 2).
img:repsd	Replicate spread — mean Cq against	No replicate groups with two or more detected wells.
img:outcomes	SOP outcomes per target	No results.
data:sop_answers	Profile answers	drawn on both sides, but no scenario moves it by more than 0 % of what it draws
data:rising_curves	Rising curves without a channel re	drawn on both sides, but no scenario moves it by more than 0 % of what it draws
data:sample_comparison	Sample comparison	drawn on both sides, but no scenario moves it by more than 0 % of what it draws

so any page in it can be reproduced

Each scenario in the demonstration set was loaded into the page itself, running from index.html in a headless browser, through the same file input and the same Read step a person uses. Once the runs were decoded, the page's own export catalogue was enumerated and every entry that reported itself ready was produced through its own figure() or text() path — the same code that writes the PDF report and the bulk archive. Figures were rasterised with the page's own svgToJpeg at twice scale; tables were taken as the CSV the page hands out.

Two sides were produced for every scenario. For a case that lives inside one run, the sound side is the normal file and the deviating side is the abnormal one. For a case that only appears across a series, the sound side is the first eight runs and the deviating side all twenty; and, because the run-level figures always draw the first run loaded, a second pair was produced from the first run alone against the last run alone.

Every catalogue entry was then compared across all forty-eight scenarios and both pairings. For a figure the measure is the share of the drawing area whose grey value differs by more than 18 of 255; for a table it is the share of shared cells that differ, with the row-count difference beside it. The pair shown on each page is the one that moved that entry most, and the other scenarios that also moved it are named at the foot of the page. An entry that no scenario moved is listed at the end with the page's own reason, quoted from the figure it drew.

All experiment files are generated. No sample, plate, operator, instrument or laboratory in them is real, and not one value in this document was measured.